Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Obtains w from $\int_0^{15} wl\,\mathrm{d}l = 1$ Allow one error if method correct	AO1.1a	M1	$\int_0^{15} wl\,\mathrm{d}l = w\left[\frac{l^2}{2}\right]_0^{15}$ $= \frac{225w}{2} = 1$
Obtains $\mathrm{E}(L^2)$ by integrating $\int l^2 \times f(l)\mathrm{d}l$ FT 'their' value for w	AO1.1a	M1	Hence $w = \frac{2}{225}$ $\mathrm{E}(L2) = \int_0^{15} l2 \times \frac{2}{225} l$
Obtains $\mathrm{E}(S)$ by evaluating $\frac{1}{15}(0+1+2+3+4+5)$ Allow one error if method correct	AO1.1a	M1	$= \frac{2}{225}\left[\frac{l^4}{4}\right]_0^{15} = \frac{225}{2}$ $\mathrm{E}(S) =$
Uses $\mathrm{E}(T) = \mathrm{E}(L^2) + \frac{1}{2}\mathrm{E}(S)$ (PI)	AO1.1a	M1	$\frac{1}{15}(0\times0+1\times1+2\times2+\ldots5\times5)$
Shows that $\mathrm{E}(T) = 114\frac{1}{3}$ Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips **AG**	AO2.1	R1	$= \frac{55}{15} = \frac{11}{3}$ $\mathrm{E}(T) = \mathrm{E}(L^2) + \frac{1}{2}\mathrm{E}(S)$ $= \frac{225}{2} + \frac{11}{6} = \frac{343}{3} = 114\frac{1}{3}$ **AG**
			Total 5 marks

Q2.

(a)

Straight line from (1 , 0.5) to (3 , ⅙).

Horizontal straight line from (3, ⅙) to (5, ⅙).

B2,1 **2**

(b) $$\mathrm{E}(X)=\frac{1}{6}\int_1^3 x(4-x)\,\mathrm{d}x+\frac{1}{6}\int_3^5 x\,\mathrm{d}x$$

*ignore limits (**both** parts attempted)*

M1

$$=\frac{1}{6}\left[2x^2-\frac{x^3}{3}\right]_1^3+\frac{1}{6}\left[\frac{x^2}{2}\right]_3^5$$

*ignore limits (**both** correct)*

A1

$$=\tfrac{1}{6}\left[(18-9)-\left(2-\tfrac{1}{3}\right)\right]+\tfrac{1}{6}\left[\tfrac{25}{2}-\tfrac{9}{2}\right]\quad=\tfrac{1}{6}\left[7\tfrac{1}{3}+8\right]$$

use of correct limits. dep on M1A1

m1

$$=2\tfrac{5}{9}$$

(AG)

A1 **4**

(c) (i) $$\mathrm{P}(X>2.5)=\tfrac{1}{3}+\tfrac{1}{2}\times\left(0.25+\tfrac{1}{6}\right)\times\tfrac{1}{2}$$

Or $$1-\int_1^{2.5}\tfrac{1}{6}(4-x)\,\mathrm{d}x=1-\left[\tfrac{1}{6}\left(4x-\frac{x^2}{2}\right)\right]_1^{2.5}$$

M1

$$=\tfrac{7}{16}$$

cao (0.4375)

A1 **2**

(ii) $$\left.\begin{aligned}\mathrm{P}(1.5<X<4.5)&=\tfrac{1}{2}\times\left(\tfrac{5}{12}+\tfrac{1}{6}\right)\times1.5\\&\quad+(4.5-3)\times\tfrac{1}{6}\end{aligned}\right\}$$

Or $$\int_{1.5}^{3}\tfrac{1}{6}(4-x)\,\mathrm{d}x+\int_3^{4.5}\tfrac{1}{6}\,\mathrm{d}x$$

M1

$= \frac{7}{16} + \frac{1}{4}$

A1

$= \frac{11}{16}$

cao (= $\frac{11}{16}$ *or 0.6875)*

A1

3

(iii) P(X > 2.5 **and** 1.5 < X < 4.5) = P(2.5 < X < 4.5)

$$\int_{2.5}^{3} \frac{1}{6}(4-x)\,dx = \left[\frac{1}{6}\left(4x - \frac{x^2}{2}\right)\right]_{2.5}^{3} = \frac{5}{48}$$

$= \frac{1}{2} \times \left(0.25 + \frac{1}{6}\right) \times 0.5 + \frac{1}{4}$

M1

$= \frac{5}{48} + \frac{1}{4}$

$= \frac{17}{48}$

cao (0.3541$\dot{6}$)

A1

2

(iv) P(X > 2.5 | 1.5 < X < 4.5) = $\frac{17/48}{11/16}$

their $\frac{(\text{iii})}{(\text{ii})}$ *iff 0 < p's < 1*

M1

$= \frac{17}{33}$

cao (allow $0.\dot{5}\dot{1}$*)*

A1

2

Alternative Solution

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{12}(x-1)(7-x) & 1 \le x < 3 \\ \frac{1}{6}(x+1) & 3 \le x < 5 \\ 1 & x \ge 5 \end{cases}$$

(i) P(X > 2.5) = 1 − F(2.5)

= 1 − $\frac{1}{12}$ (2.5 − 1)(7 − 2.5)

(M1)

= 1 − $\frac{1}{12}$ × 1.5 × 4.5

= 1 − 0.5625

= 0.4375 **or** $\frac{7}{16}$

cao

(A1)

(2)

(ii) P(1.5 < X < 4.5) = F(4.5) − F(1.5)

$= \frac{1}{6}(4.5 + 1) - \frac{1}{12}$ (1.5 − 1)(7 − 1.5)

(M1)

$= \frac{11}{12} - \frac{11}{48}$

(A1)

$= \frac{11}{16}$ **or** 0.6875

cao

(A1)

(3)

(iii) P(X > 2.5 **and** 1.5 < X < 4.5)

= P(2.5 < X < 4.5)

= F(4.5) − F(2.5)

$= \frac{11}{12} - \frac{9}{16}$

(M1)

$= \frac{17}{48}$

cao

(A1)

(2)

(iv) P(X > 2.5 | 1.5 < X < 4.5)

(M1)

$$= \frac{F(4.5) - F(2.5)}{F(4.5) - F(1.5)} \text{ or } \frac{\text{their (iii)}}{\text{their (ii)}}$$

$$= \frac{17/48}{11/16}$$

$= \frac{17}{33}$ **or** (allow $0.\dot{5}\dot{1}$)

cao

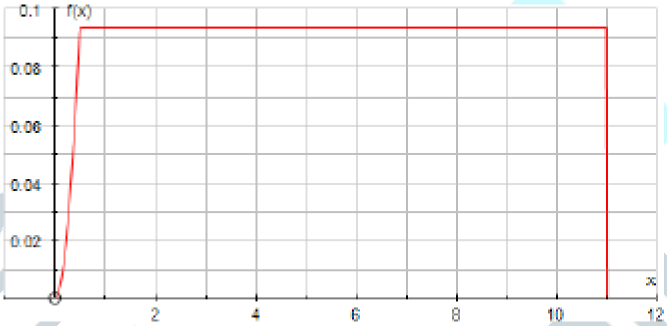
(A1)

(2)

[15]

Q3.

(a)



B1 for concave curve from

(0 , 1) to $\left(\frac{1}{2}, \frac{3}{32}\right)$

B1 for horizontal straight line

f = $\frac{3}{32}$ *from* $\left(\frac{1}{2}, \frac{3}{32}\right)$ *to* $\left(\frac{1}{2}, \frac{3}{32}\right)$

B1 for correct axes

B3

3

(b) (i) $P\left(X \geq 8\frac{1}{3}\right) = \left[\frac{3}{32} \times \left(11 - 8\frac{1}{3}\right)\right]$

$$= \frac{3}{32} \times \frac{8}{3}$$

Any correct method attempted in either part

M1

$$= \frac{1}{4}$$

AG

A1

(ii) $P(X \geq 3) = \frac{3}{32} \times (11 - 3)$

$= \frac{3}{4}$

Any correct method attempted
AG

A1

3

(c) (i) Interquartile Range = $5\frac{1}{3}$
cao

B1

(ii) Median = $5\frac{2}{3}$
Cao

B2

Alternative:

$\frac{1}{64} + \frac{3}{32}\left(m - \frac{1}{2}\right) = \frac{1}{2}$

$\Rightarrow 3\left(m - \frac{1}{2}\right) = 15.5 \Rightarrow m = 5\frac{2}{3}$

***sc** if B0 then:*
M1 for correct method seen

$\frac{1}{2}\left(8\frac{1}{3} + 3\right)$ **or** $\frac{1}{2} \times 11\frac{1}{3}$

or $\frac{3}{32}(11 - m) = \frac{1}{2} \Rightarrow 11 - 5\frac{1}{3}$

3

(d) $P[(X < m) \cap (X \geq 3)] = \frac{1}{4}$

$\left(\frac{3}{4} - \frac{1}{2}\right)$ ***attempted***

B1

$P(X < m \mid (X \geq 3) = \frac{1/4}{3/4} = \frac{1}{3}$

(their p) $\bigg/ 3 \Big/ 4$ *for 0 < p < 1*

M1

cao

A1

Alternative:

(Ratio of relevant two areas)

P(X < m | X ≥ 3) = $\dfrac{2\frac{2}{3}}{8} = \dfrac{1}{3}$

cao

3

[12]

Q4.

(a) $$\mathrm{F}(x) = \int \frac{3}{8}(x^2+1)\mathrm{d}x$$

Ignore limits

M1

$$= \frac{3}{8}\left[\frac{x^3}{3} + x\right] \text{ or } = \frac{1}{8}x^3 + \frac{3}{8}x$$

Either

A1

$$= \frac{1}{8}x(x^3+3)$$

*(including use of correct limits 0 and x **or** +c used and evaluated) (AG)*

A1

3

(b) $$\mathrm{F}(m) = \frac{1}{2}$$

B1

$$\mathrm{F}(1) = \frac{1}{8} \times 1 \times 4 = \frac{1}{2}$$

AG

B1

2

(c) Upper quartile lies in range 1 < x < 2 such that $\mathrm{F}(q) = \dfrac{3}{4}$

$$\frac{1}{2}+\int_1^q \frac{1}{4}(5-2x)dx=\frac{3}{4}$$

$$\int_1^q \frac{1}{4}(5-2x)dx=\frac{1}{4}$$

Alternative: $$\int_q^2 \frac{1}{4}(5-2x)\mathrm{d}x=\frac{1}{4}$$

M1

$$\left[5x-x^2\right]_1^q=1$$

$$\left[5x-x^2\right]_q^2=1$$

$5q - q^2 - 4 = 1$

$(10 - 4) - (5q - q^2) = 1$

$6 - 5q + q^2 = 1$

$q^2 - 5q + 5 = 0$

$q^2 - 5q + 5 = 0$

A1

$$q=\frac{5\pm\sqrt{25-20}}{2} \quad \text{or} \quad \frac{1}{2}\left(5\pm\sqrt{5}\right)$$

Correct use of formula (OE) to give the two surd answers to given quadratic equation

M1

but 1 q 2 [or (q 2)]

m1

$$\therefore\ q=\frac{1}{2}\left(5-\sqrt{5}\right)$$

Must qualify with a numerical comparison, not just quote the given answer; dep on M1; AG

A1 5

(d) $$\mathrm{P}(X>1.5)=\frac{1}{2}\left[\frac{1}{2}+\frac{1}{4}\right]\times\frac{1}{2}$$

$$\mathrm{P}(X<1.5)=0.5+\frac{1}{2}\left[\frac{3}{4}+\frac{1}{2}\right]\times\frac{1}{2}\quad \text{(M1)}$$

M1

$$=\frac{3}{16}(0.1875)$$

$$= \frac{1}{2} + \frac{1}{2} \times \frac{5}{4} \times \frac{1}{2}$$

$$= \frac{1}{2} + \frac{5}{16} = \frac{13}{16} \qquad \text{(A1)}$$

A1

$$\text{P}(X > q) = \frac{1}{4}(0.25)$$

$$\text{P}(X < q) = \frac{3}{4}(0.75) \qquad \text{(B1)}$$

B1

$$\text{P}(q < X < 1.5) = \frac{1}{4} - \frac{3}{16}$$

$$= \frac{1}{16} \quad (0.0625)$$

$$\text{P}(q < X < 1.5) = \frac{13}{16} - \frac{3}{4} = \frac{1}{16} \qquad \text{(A1)}$$

(0.0625)

A1

OR

$$\int_{1\frac{1}{2}}^{2} \frac{1}{4}(5 - 2x)\mathrm{d}x = \frac{3}{16} \text{ etc} \quad \text{(M1A1)}$$

OR

$$\int_{q}^{1.5} \frac{1}{4}(5 - 2x)\mathrm{d}x = \frac{1}{4}\left[5x - x^2\right]_q^{1.5} \qquad \text{(M1)}$$

(correct integration and limits)

Allow use of q = 1.38 to q = 1.382 in limits for M1

*Whatever follows **must** be **exact***

$$= \frac{1}{4}\left[(7.5 - 2.25) - (5q - q^2)\right] \qquad \text{(A1)}$$

for use of $5q - q^2 = 5$ ***or*** *showing*

$5q - q^2 = 5$ *by substituting* $q = \frac{1}{2}\left(5 - \sqrt{5}\right)$ *(A1)*

$$= \frac{1}{4}[5.25 - 5] = \frac{1}{16} \qquad \text{(A1)}$$

NB statement $\text{F}(1.5) - \frac{3}{4} = \frac{1}{16}$ (OE) scores 4 marks

4

Alternative:

$$\int_q^{1.5} \frac{1}{4}(5-2x)\mathrm{d}x = \left[-\frac{1}{16}(5-2x)^2\right]_{\frac{5-\sqrt{5}}{2}}^{1.5}$$ (M1)

$$= -\frac{1}{16}(4) - \left[-\frac{1}{16}\left(\sqrt{5}\right)^2\right] \text{ (sub)}$$ (A1)

$$= -\frac{4}{16} + \frac{5}{16}$$ (A1)

$$= \frac{1}{16}$$ (A1)

Alternative using F(x):

for 1 ≤ x ≤ 2

$$\mathrm{F}(x) = \frac{1}{2} + \int_1^x \frac{1}{4}(5-2x)dx$$

$$= \frac{1}{2} + \frac{1}{4}\left[5x - x^2\right]_1^x$$

$$= \frac{1}{2} + \frac{1}{4}\left[\left(5x - x^2\right) - (5-1)\right]$$

$$= \frac{1}{4}\left(2 + 5x - x^2 - 4\right)$$

$$= \frac{1}{4}\left(5x - x^2 - 2\right) \text{ (seen or used)}$$ (M1)

$$\mathrm{F}(1.5) = \frac{1}{4}(7.5 - 2.25 - 2) = \frac{3.25}{4}$$

$$= 0.8125 = \frac{13}{16}$$ (A1)

$$\mathrm{F}(q) = \frac{1}{16}\left(50 - 10\sqrt{5} - \left(25 - 10\sqrt{5} + 5\right) - 8\right)$$

$$= \frac{12}{16} \text{ OE}$$ (B1)

$$\mathrm{P}(q < X < 1.5) = \frac{13}{16} - \frac{12}{16} = \frac{1}{16}$$ (A1)

[14]

Q5.

(a) (i) $\mathrm{E}(T) = \frac{1}{2}(25 + -5) = 10$

CAO

B1 **1**

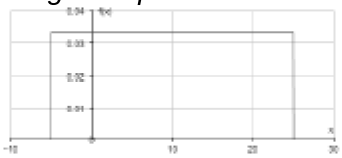
(ii) $\mathrm{Var}(T) = \frac{1}{12}(25 - -5)^2 = 75$

CAO

B1 **1**

(b) $P(-2 < T < 2) = \frac{2}{15}$ (OE)

Diagram optional



B1

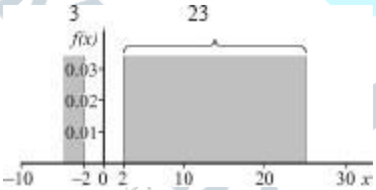
P(magnitude at least 2 minutes)
$= 1 - P(-2 < T < 2)$
$= 1 - \frac{4}{30}$

M1

$= \frac{13}{15}$ (OE) = 0.867
CAO (AWRT)

A1

or



$\frac{1}{30}(3+23) = \frac{26}{30} = \frac{13}{15}$

Alternative

$P(T > 2) = \frac{23}{30}\ (0.76\dot{6})$

or $P(T < -2) = \frac{1}{10}$

B1

or

$\int_{-5}^{-2} \frac{1}{30}\,dt + \int_{2}^{25} \frac{1}{30}\,dt = \frac{1}{10} + \frac{23}{30} = \frac{13}{15}$

P(magnitude at least 2 minutes)
$= P(T < -2) + P(T > 2)$
$= \frac{13}{15}$ *for M1 A1*

or

$$1-\int_{-2}^{2}\frac{1}{30}\mathrm{d}t = 1-\left[\frac{t}{30}\right]_{-2}^{2}$$

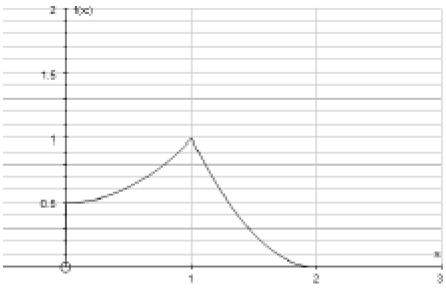
$$= 1-\frac{4}{30} = \frac{26}{30} = \frac{13}{15}$$

3

[5]

Q6.

(a)



B1 for axes
B1 for curve from (0, 0.5) to (1, 1)
B1 for curve from (1, 1) to (2, 0)

B3

3

(b) $$\mathrm{P}(X \le 1) = \int_0^1 \tfrac{1}{2}(x^2+1)\,\mathrm{d}x$$

M1

$$= \left[\frac{x^3}{6}+\frac{x}{2}\right]_0^1$$

A1

$$= \left[\frac{1}{6}+\frac{1}{2}\right] = \frac{2}{3}$$

0.667

A1

3

(c) $$\mathrm{E}(X^2) = \int_0^1 x^2 \times \frac{1}{2}(x^2+1)dx + \int_1^2 x^2(x-2)^2\,dx$$

both integrals seen

M1

$$= \left[\frac{x^5}{10} + \frac{x^3}{6}\right]_{x=0}^{x=1} + \left[\frac{x^5}{5} - x^4 + \frac{4x^3}{3}\right]_{x=1}^{x=2}$$

A1A1

$$= \left(\frac{1}{10} + \frac{1}{6}\right) + \left(\left[\frac{32}{5} - 16 + \frac{32}{3}\right] - \left[\frac{1}{5} - 1 + \frac{4}{3}\right]\right)$$

dep(M1)

m1

$$= \frac{4}{5}$$

AG

A1

5

(d) (i) E(X) = $\frac{19}{24}$ and kVar(X) = 499

Var(X) = E(X^2) – E^2(X)

$$= \frac{4}{5} - \left(\frac{19}{24}\right)^2$$

M1

$$= \frac{499}{2880}$$ (0.173)

A1

$\Rightarrow$ k = 2880

CAO

A1

3

(ii) E($5X^2 + 24X - 3$)

= 5E(X^2) + 24E(X) – 3

M1

$$= 5 \times \frac{4}{5} + 24 \times \frac{19}{24} - 3$$

= 20

CAO

A1

2

(iii) Var($12X - 5$) = 144Var(X)

M1

$= 144 \times \frac{499}{2880}$

$= \frac{499}{20}$ or (24.95)

CAO (AWFW 24.9 to 25)

A1

2

[18]

Q7.

(a) Median = 1

B1

Lower quartile = $\frac{1}{2}$

B1

2

(b) F(1) = $\frac{1}{2}$

For 1 ≤ *x* ≤ 4

$\int \frac{1}{18}(x-4)^2\,dx$

ignore limits

M1

$= \left[\frac{1}{54}(x-4)^3\right]_1^x$

$= \left[\frac{1}{54}(x-4)^3 + \frac{1}{2}\right]$

Correct integration + correct limits seen or used

A1

$F(x) = \left[\frac{1}{54}(x-4)^3 + \frac{1}{2}\right] + \frac{1}{2}$

adding $\frac{1}{2}$ ***or*** *F(1)*

m1

$= 1 + \frac{1}{54}(x-4)^3$

CAO (AG)

A1

Alternative

$\int \frac{1}{18}(x-4)^2\,dx = \frac{1}{54}(x-4)^3 + c$

(M1)

F(1) = $\frac{1}{2}$ $\Rightarrow$ $c = 1$

(m1)(A1)

F(x) = 1 + $\frac{1}{54}$ $(x-4)^3$

(A1) 4

Alternative

$\int \frac{1}{18}(x-4)^2\,dx$ (M1)

$= \int_1^x \frac{1}{18}(x^2 - 8x + 16)\,dx$

$= \frac{1}{18}\left[\frac{x^3}{3} - 4x^2 + 16x\right]_1^x$ (A1)

$F(x) = \frac{1}{2} + \frac{1}{54}[x^3 - 12x^2 + 48x]_1^x$ (m1)

$= \frac{1}{2} + \frac{1}{54}(x^3 - 12x^2 + 48x - 37)$

$= 1 + \frac{1}{54}(x^3 - 12x^2 + 48x - 64)$

$= 1 + \frac{1}{54}(x-4)^3$ (A1)

(c) P(2 ≤ X ≤ 3) = $\frac{53}{54} - \frac{46}{54}$

F(3) – F(2)

M1

$= \frac{7}{54}$ (0.130)

$0.1\dot{2}9\dot{6}$

A1 2

(d) (i) $F(q) = \frac{3}{4}$

***use** of F(q) =* $\frac{3}{4}$

M1

$1 + \frac{1}{54}(q-4)^3 = \frac{3}{4}$

$\frac{1}{54}(q-4)^3 = -\frac{1}{4}$

(either)

M1

(×54) $\Rightarrow (q-4)^3 = -13.5$

AG

A1

3

(ii) $q - 4 = \sqrt[3]{-13.5} = -2.3811 \quad q = 1.619$ (3dp)

CAO

B1

1

[12]

Q8.

(a) $P\left(-\frac{3c}{4} < X < \frac{3c}{4}\right)$

$= \frac{\frac{3c}{4} + c}{4c} - \frac{\frac{-3c}{4} + c}{4c}$

or $= \frac{3c}{2} \times \frac{1}{4c}$

M1

$= \frac{6c}{16c}$

$= \frac{3}{8}$ or 0.375

CAO

A1

2

(b) For $-c \le x \le 3c$

$$f(x) = \frac{d}{dx}\left(\frac{x+c}{4c}\right)$$

use of f(x) = F′(x)

M1

$$= \frac{1}{4c}$$

For $x > 3c$ and $x < -c$

$$f(x) = \frac{d}{dx}(F) = 0$$

for $\frac{1}{4c}$ *and 0*

A1

2

(c) (i) Rectangular distribution:

$$E(X) = \frac{1}{2}(-c + 3c) = c$$

B1

1

(ii) Var $(X) = \frac{1}{12}(3c - -c)^2 = \frac{4c^2}{3}$

Allow $\frac{16c^2}{12}$

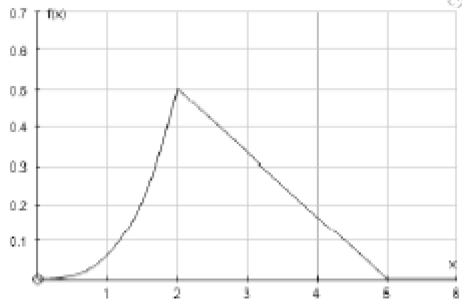
B1

1

[6]

Q9.

(a)



for concave curve from (0, 0) to (2, 0.5)

B1

for straight line from (2, 0.5) to (5, 0)

B1

for axes [2, 5; 0.5] seen

B1

3

(b) $P(X \geq 2) = \frac{1}{2} \times 3 \times 0.5 = 0.75$

$\Rightarrow$ F(2) = 0.25

***Alternative**:*

$\int \frac{1}{6}(5-x)dx = \frac{1}{6} \times \frac{(5-x)^2 \times (-1)}{2}$

$= \frac{1}{12}(5-x)^2$

$2 \leq x \leq 5$

$F(x) = F(2) + \int_2^x \frac{1}{6}(5-x)dx$

$= 0.25 + \frac{1}{6}\left[5x - \frac{x^2}{2}\right]_2^x$

M1

$= 0.25 + \frac{1}{6}\left(5x - \frac{x^2}{2}\right) - \frac{1}{6}(10-2)$

A1

$= 0.25 - \frac{8}{6} + \frac{5x}{6} - \frac{x^2}{12}$

$= -\frac{1}{12}(x^2 - 10x + 13)$

M1

$= 1 - \frac{1}{12}(5-x)^2$

A1

Or

F(x) = 1 – Area Δ (base x, 5)

= 1 – $\frac{1}{2}(5-x)\frac{1}{6}(5-x)$

= 1 – $\frac{1}{12}(5-x)^2$

4

(c) $P(X \leq 4) = F(4)$

$= 1 - \frac{1}{12}(5-4)^2 = \frac{11}{12}$ (0.916 to 0.917)

B1

$F(3) = 1 - \frac{1}{12}(2)^2 = \frac{2}{3}$ (0.667)

B1

$P(X \geq 3 \text{ and } X \leq 4) = F(4) - F(3)$

$= \frac{11}{12} - \frac{2}{3} = \frac{1}{4}$ (0.25)

B1

$P(X \geq 3 | X \leq 4) = \frac{F(4) - F(3)}{F(4)}$

M1

$= \frac{1/4}{11/12} = \frac{3}{11}$

A1

Alternative:

$P(X \geq 3 \mid X \leq 4)$

$= \frac{F(4) - F(3)}{F(4)}$ (M1)

$= 1 - \frac{F(3)}{F(4)}$

$= 1 - \frac{2/3}{11/12}$

$= 1 - \frac{8}{11}$ (B1) ($0.7\dot{2}\dot{7}$)

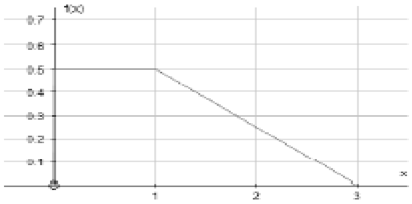
$= \frac{3}{11}$ *(AWFW 0.272 to 0.273)*

5

[12]

Q10.

(a) Sketch:



1 for straight line 0 ≤ x ≤ 1 from (0, 0.5) to (1, 0.5)

1 for straight line 1 ≤ x ≤ 3 from (1, 0.5) to (3, 0)

1 for axes [must have at least (0, 0.5)
(1, 0) and (3, 0) labelled]

B3

3

(b) P($X \le \eta$) = F(η) = 0.5

M1

($\Rightarrow$ η = 1 (from graph))
AG

A1

2

(c) $$\mu = \mathrm{E}(X) = \int_0^1 \left(\frac{x}{2}\right) \mathrm{d}x + \int_1^3 x\left(\frac{3-x}{4}\right) \mathrm{d}x$$

Both integrals stated

M1

$$= \left[\frac{x^2}{4}\right]_0^1 + \frac{1}{4}\left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_1^3$$

Either

A1

$$= \frac{1}{4} + \frac{1}{4}\left[\left(\frac{27}{2} - 9\right) - \left(\frac{3}{2} - \frac{1}{3}\right)\right]$$

Correct limits used on both integrals +combined dep M1

m1

$$= \frac{1}{4} + \frac{5}{6} \qquad (0.25 + 0.83\dot{3})$$

$$= 1\frac{1}{12}$$

(CAO)

A1

(d) Area of Δ

$$= P\left(X > 2\frac{1}{4}\right) = \frac{1}{2} \times \frac{3}{4} \times \frac{3 - 2\frac{1}{4}}{4}$$

M1ft

$$= \frac{3}{32} \times \frac{3}{4} = \frac{9}{128}$$

$$\therefore P\left(X < 2\frac{1}{4}\right) = 1 - \frac{9}{128}$$

M1ft

$$= \frac{119}{128} (0.9296875)$$

A1

Alternative:

For $1 \le x \le 3$

$F(x) = 1 - \frac{1}{8}(3 - x)^2$ *M1ft*

$$\Downarrow$$

$$F\left(2\frac{1}{4}\right) = 1 - \frac{1}{8} \times \frac{9}{16}$$ *M1ft*

$$= \frac{119}{128}$$

CAO

or

$$\int_{2\frac{1}{4}}^{3} \frac{3 - x}{4} \, dx \left(= \frac{9}{128}\right)$$

M1ft

$$= 1 \int_{2\frac{1}{4}}^{3} \frac{3 - x}{4} \, dx$$

M1ft

$$= 1 - \frac{1}{4}\left[3x - \frac{x^2}{2}\right]_{2\frac{1}{4}}^{3}$$

$$= 1 - \frac{1}{4}\left[9 - \frac{9}{2} - \frac{27}{4} + \frac{81}{32}\right]$$

$$= 1 - \frac{1}{4} \times \frac{9}{32} = \frac{119}{128}$$

A1

or (1 – 0.0703125 = 0.9296875)

Alternative

$$f\left(2\frac{1}{4}\right) = \frac{3}{16} = 0.1875$$

$$P(X < 3\mu - \eta) = P\left(X < 2\frac{1}{4}\right)$$

$$= \frac{1}{2} + \frac{1}{2}\left(\frac{3}{16} + \frac{1}{2}\right) \times 1\frac{1}{4}$$ *M1ft*

$$= \frac{1}{2} + \frac{55}{128}(0.4296875)$$ *M1ft*

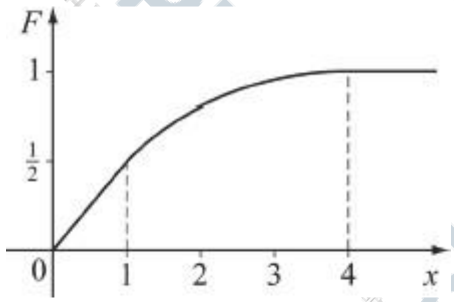
$$= \frac{119}{128}(0.930)$$ *A1*

3

[12]

Q11.

(a) (i)



B1 for axes 0 to 4 & 0 to 1

B1 for straight line 0 – $\left(1, \frac{1}{2}\right)$

B1 for convex curve from $\left(1, \frac{1}{2}\right)$ *to (4,1)*

B1 for at least the straight line for x > 4

B4

4

(ii) F(q_1) = 0.25 $\Rightarrow \frac{1}{2}q_1 = 0.25$

From sketch, or from F(x), Median = 1.

M1

$\Rightarrow q_1 = \frac{1}{2}$

$\frac{1}{2}x$ *is linear on (0, 0) to* $\left(1, \frac{1}{2}\right)$

$\therefore q_1 = \frac{1}{2}$ ***AG***

A1

2

(iii) F(1.6) = 0.744
F(1.7) = 0.775

M1

$F(q_3) = 0.75$

M1

$\Rightarrow 1.6 < q_3 < 1.7$
AG

A1

3

(b) (i) f(x) = F(x)

M1

$\Rightarrow f(x) = \frac{1}{2}$ for $0 \le x \le 1$

A1

$\Rightarrow a = \frac{1}{2}$

f(1) = α = 9β *B1*

$\int_{-\infty}^{\infty} f(x)dx = 1 \Rightarrow$

$[ax]_0^1 + \left[\frac{\beta(x-4)^3}{3}\right]_1^4 = 1$ *M1*

$\Rightarrow$ for $1 \le x \le 4$
$f(x) = \frac{1}{54}(3x^2 - 24x + 48)$

A1

$= \frac{3}{54}\left(x^2 - 8x + 16\right)$

M1

$= \frac{1}{18}(x-4)^2$

$\Rightarrow \beta = \frac{1}{18}$

A1

$\Rightarrow$ ***α* + 9*β* = 1** *A1*

Solving: *M1*

$a = \frac{1}{2}$ and $\beta = \frac{1}{18}$ A1

5

(ii) $\mathrm{E}(X) = \int_0^1 \frac{1}{2}x\,dx + \int_1^4 \frac{1}{18}\left(x^3 - 8x^2 + 16x\right)dx$

Both seen

M1

$= \left[\frac{1}{4}x^2\right]_0^1 + \frac{1}{18}\left[\frac{x^4}{4} - \frac{8x^3}{3} + 8x^2\right]_1^4$

A1A1

$= \frac{1}{4} + \frac{7}{8}$

Dependent on M1

m1

CAO

A1

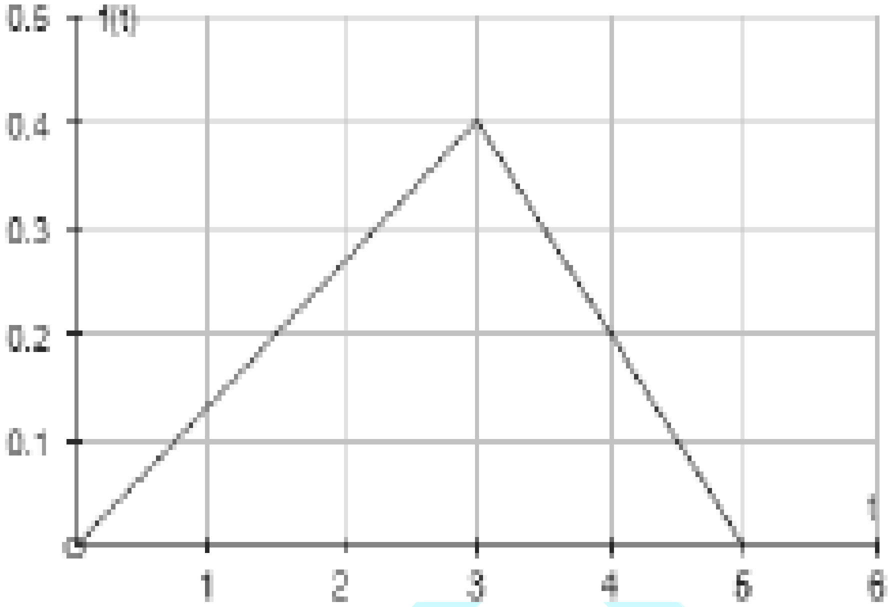
$= 1\frac{1}{8}$

5

[19]

Q12.

(a)



line segment on 0 −3

B1

line segment on 3 −5

B1

scales (0 − 0.4 vertical; 0 – 5 horizontal)

B1

3

(b) (i) $P(T \le 2) = \frac{1}{2} \times 2 \times \frac{4}{15}$

M1

$= \frac{4}{15}$

(0.267)

A1

2

(ii) P(2 < T < 4) = 1 – (P(T < 2) + P(T > 4))

M1

$= 1 - \left(\frac{4}{15} + \frac{1}{2} \times \frac{1}{5}\right)$

for $P(T > 4) = \frac{1}{10}$

$\frac{1}{2}d[(f_2 + f_4) + 2f_3]$

$f_2 = \frac{4}{15}; f_4 = \frac{1}{5}; f_3 = \frac{2}{5}; d = 1$

A1

$= 1 - \frac{4}{15} - \frac{1}{10}$

$= \frac{19}{30}$

(0.633)

A1

3

(c) $E(T) = \int_0^3 \frac{2}{15}t^2 \, dt = \int_3^5 t\left(1 - \frac{1}{5}t\right) dt$

both integrals seen

M1

$= \left[\frac{2}{45}t^3\right]_0^3 + \left[\frac{1}{2}t^2 - \frac{1}{15}t^3\right]_3^5$

B1B1

$= \frac{6}{5} + \frac{25}{6} - \frac{27}{10}$

$= 2\frac{2}{3}$

OE

A1

4

[12]

Q13.

(a) P(X < 0) = F(0)

M1

$= \dfrac{1}{k+1}$

A1 **2**

(b) $(q_1 + 1) \times \dfrac{1}{k+1} = \dfrac{1}{4}$

alternative (from a sketch)

M1

$q_1 + 1 = \dfrac{1}{4}(k+1)$

A1

$q_1 = \dfrac{1}{4}(k+1) - 1$

OE

A1 **3**

(a) $f(x) = \dfrac{d}{dx}(F(x))$

use of

M1

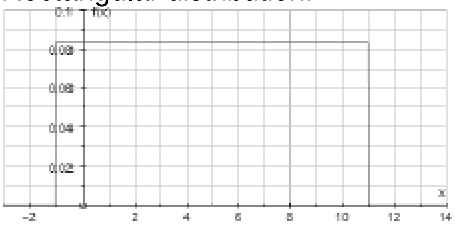
$$\left.\begin{array}{ll} = \dfrac{1}{k+1} \times \dfrac{d}{dx}(x+1) & \\ = \dfrac{1}{k+1} & -1 \le x \le k \\ = 0 & \text{otherwise} \end{array}\right\}$$

AG; $\dfrac{1}{k+1}$ *clearly deduced*

A1 **2**

$k = 11 \Rightarrow f(x) = \begin{cases} \frac{1}{12} & -1 \le x \le 11 \\ 0 & \text{otherwise} \end{cases}$

(b) (i) Rectangular distribution:



horizontal line on [–1, 11]

B1

at $f = \frac{1}{12}$

B1

2

$\mathrm{E}(X) = \frac{1}{2}(-1 + 11) = 5$

(ii)

B1

$\mathrm{Var}(X) = \frac{1}{12}(11 - -1)^2 = 12$

B1

2

$\mathrm{P}(q_1 < X < \mathrm{E}(X)) = \mathrm{P}(2 < X < 5)$

(iii)

$= (5 - 2) \times \frac{1}{12}$

M1

= 0.25

AG

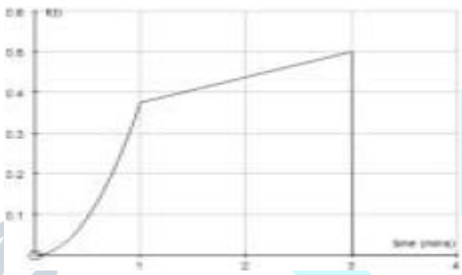
A1

2

[13]

Q14.

(a)



for curve

B1

for line

B1

for axes

B1

3

(b) $P(T \geq 1) = \frac{1}{2} \times \frac{7}{8} \times 2 = \frac{7}{8}$

OE

M1A1

2

(c) (i) For $1 \leq t \leq 3$

$\int_1^t \frac{1}{16}(t+5)\,dt = \left[\frac{1}{32}t^2 + \frac{5}{16}t\right]_1^t$

M1A1

$F(1) = \frac{1}{8}$

B1

$$\mathrm{F}(t) = \frac{1}{8} + \frac{1}{32}t^2 + \frac{5}{16}t - \frac{11}{32}$$

Use of : $F(t) = F(1) + \int_1^t \frac{1}{16}(t+5)\,\mathrm{d}t$

M1

$$\mathrm{F}(t) = \frac{1}{32}(t^2 + 10t - 7)$$

AG

A1

5

Alternative:

$$\int \tfrac{1}{16}(t+5)\,\mathrm{d}t$$

$$= \frac{1}{16}\left(\frac{1}{2}t^2 + 5t + c\right)$$

(M1)(A1)

$$\mathrm{F}(1) = \frac{1}{8}$$

(B1)

$\Rightarrow c = -3.5$

(M1)

$$\mathrm{F}(t) = \frac{1}{32}(t^2 + 10t - 7)$$

(A1)

(ii) $\frac{1}{32}(m^2 + 10m - 7) = 0.5$

M1

$m^2 + 10m - 23 = 0$

A1

$$m = \frac{-10 \pm \sqrt{192}}{2} = -5 \pm \sqrt{48}$$

(or any valid method)

m1

$= -5 \pm 4\sqrt{3}$

$(m > 0)$

$m = 4\sqrt{3} - 5 = 1.93$

(1.9282)

A1

4

[14]

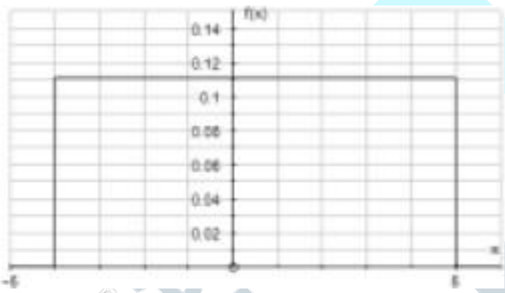
Q15.

(a) $f(x) = \begin{cases} \frac{1}{9} & -4 \le x \le 5 \\ 0 & \text{otherwise} \end{cases}$

M1A1

2

(b)



horizontal line from –4 to 5

B1

for drawn at $\frac{1}{9}$

B1

2

(c) $P(X > 2) = \frac{1}{9} \times 3$

F(5) – F(2)

$= 1 - \frac{2}{3}$

M1

$= \frac{1}{3}$

A1

2

(d) Mean = $\frac{1}{2}$

B1

Variance = $\frac{1}{12} \times 81$

= 6.75

B1

2

[8]

Q16.

(a) For a Rectangular Distribution

$$f(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

(–0.05, 0.05) $\Rightarrow$

(explain error ± 0.05)

B1

$$\frac{1}{b-a} = \frac{1}{0.05-(-0.05)} = \frac{1}{0.1} = 10$$

M1A1

(Area = 10 × 0.1 =1)

3

(b) P(–0.01 < X < 0.02) = 0.03 × 10 = 0.3

M1A1

2

(c) Mean = 0

CAO

B1

Standard deviation = 0.0289

$\frac{1}{20\sqrt{3}}$ *OE*

B1

2

[7]

Q17.

(a) (i) Area = *k*(*b* – *a*) = l

$\Rightarrow \quad k = \dfrac{1}{b-a}$

AG

E1 **1**

(ii) $E(X) = \int_a^b kx \, dx$

M1

$= \left(\dfrac{kx^2}{2}\right)\Big|_a^b$

A1

$= \dfrac{1}{2} k (b^2 - a^2)$

$\dfrac{1}{2} \times \dfrac{1}{(b-a)} \times (b-a)(a+b)$

(factors shown)

M1

$= \dfrac{1}{2}(a + b)$

AG

A1 **4**

(b) (i) $\mu = 1$

B1 **1**

(ii) $\sigma^2 = \text{Var}(X) = \dfrac{1}{12}(b-a)^2$

$= \dfrac{1}{12} \times 6^2$

M1

$= 3$

$\therefore \ \sigma = \sqrt{3}$

AWRT 1.73

A1 **2**

(iii) $P\left(X < \frac{2-\mu}{\sigma}\right) = P\left(X < \frac{1}{\sqrt{3}}\right)$

(on their μ and σ)

M1ft

$= \frac{1}{6} \times 2.577$

M1ft

= 0.430

A1

3

[11]

Q18.

(a) $\alpha = \frac{1}{0.4-(-0.4)} = 1.25$

B1

1

(b) $E(R) = \frac{1}{2}(-0.4+0.4) = 0$

B1

$Var(R) = \frac{1}{12}(0.8)^2 = \frac{4}{75}$

M1

Standard deviation $= \frac{2}{5\sqrt{3}} = 0.231$

A1

3

(c) P(|R| < 3) = P (–0.3 < R < 0.3)

(or seen on diagram)

M1

= 0.6 × 1.25

= 0.75

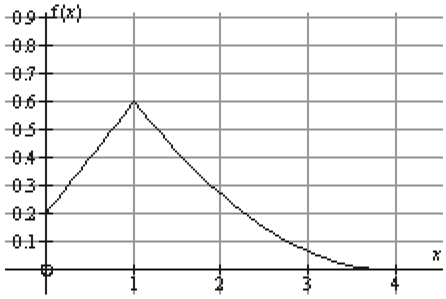
A1

2

[6]

Q19.

(a)



B1 for line segment
(0, 0.2) to (1, 0.6)
B1 for correct shaped curve
(1, 0.6) to (4 ,0)

B2 **2**

(b) (i) for $0 \le x \le 1$

$$F(x) = \int_0^x \frac{1}{5}(2x)\,dx$$

M1

$$= \left.\frac{1}{5}(x^2 + x)\right|_0^x$$

A1

$$\frac{1}{5}x(x+1)$$

A1 **3**

(ii) $P(X \ge 0.5) = 1 - F(0.5)$

$$= 1 - \frac{1}{5} \times 0.5 \times 1.5$$

M1

$$= 1 - 0.15$$
$$= 0.85$$

(OE)

A1 **2**

(iii) $F(q_1) = 0.25$

$\frac{1}{5}x(x+1) = 0.25$

$\downarrow$

B1

$F(0.5) = \frac{3}{20} = 0.15 < 0.25$

$F(0.75) = \frac{21}{80} = 0.2625 > 0.25$

Attempt to solve

$\downarrow$

$q_1 = 0.725$

$\Downarrow$

M1

$\therefore\ 0.5 < q_1 < 0.75$

$0.5 < q_1 < 0.75$ *AG*

A1

3

[10]

Q20.

(a) (i) $E(X) = \frac{1}{2}b$

B1

1

(ii) $E(X^2) = \int_0^b \frac{1}{b}x^2\,dx$

M1

$= \frac{1}{b}\left[\frac{x^3}{3}\right]_0^b$

For correct integration

A1

$= \frac{1}{b}\left(\frac{b^3}{3}\right)$

$= \frac{1}{3}b^2$

OE

A1

$\text{Var}(X) = \frac{1}{3}b^2 - \left(\frac{b}{2}\right)^2$

Depending on using integration to get $E\left(X^2\right)$

m1

$= \frac{1}{3}b^2 - \frac{1}{4}b^2$

$= \frac{1}{12}b^2$

AG

A1

5

(b) P(|T| > 0.02) = 1 – P (–0.02 < *T* < 0.02)

M1

= 1 – 0.04 × 5

M1

= 0.8

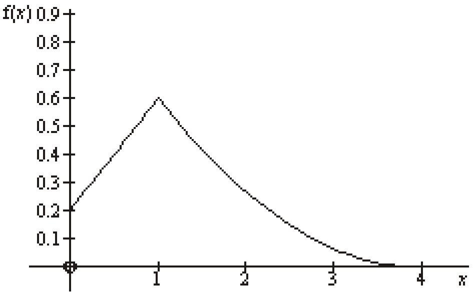
A1

3

[9]

Q21.

(a)



B1 for line segment (0,0.2) to (1,0.6)
B1 for correctly shaped curve (1,0.6) to (4,0)

B2

2

(b) (i) for $0 \le x \le 1$

$$F(x) = \int_0^x \frac{1}{5}(2x+1)\,dx$$

Ignore limits

M1

$$= \left[\frac{1}{5}(x^2 + x)\right]_0^x$$

Ignore limits

A1

$$= \frac{1}{5}x(x+1)$$

A1

3

(ii) $$P(X \le 1) = F(1)$$
$$= \frac{2}{5}$$

B1

1

(iii) $$P(X \ge x) = \frac{17}{20} \quad \Rightarrow \quad F(x) = \frac{3}{20}$$

M1

$\frac{1}{5}x(x+1) = \frac{3}{20}$

m1

$x(x+1) = \frac{3}{4}$

$x^2 + x - \frac{3}{4} = 0$

A1

$\left(x - \frac{1}{2}\right)\left(x + \frac{3}{2}\right) = 0$

Any valid method attempted

m1

$x - \frac{1}{2}$

A1

5

(iv) Since $F(1) = 0.4$, q lies in $0 \le r \le 1$

$F(q) = \frac{1}{5}(q^2 + q) = 0.25$

M1

$\Rightarrow \quad q^2 + q = 1.25$

$q^2 + q - 1.25 = 0$

A1

$\Rightarrow \quad q = \frac{-1 \pm \sqrt{1 - 4 \times (-1.25)}}{2}$

m1

$q = \frac{1}{2}\left(\sqrt{6} - 1\right) \quad (q > 0)$

AWFW (0.724 to 0.725)

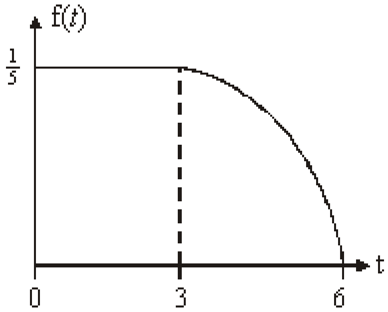
A1

4

[15]

Q22.

(a)



B1 2 axes with scales
B1 Horizontal line at 0.2 from 0 to 3
B1 Curve from 3 to 6

B3

3

(b) P (T = 3) = 0

B1

1

(c) P ($T \geq 3$) = 1 – P (T < 3)

M1

$= 1 - \frac{3}{5}$

$= \frac{2}{5}$

$$\int_3^6 \frac{1}{45} t(6-t)\,dt = \frac{2}{5}$$

A1

2

(d) $\int_0^m \frac{1}{5}\,dt$ = 0.5

P ($T \leq 3$) = 0.6

M1

$\therefore \left(\frac{t}{5}\right)_0^m = 0.5$

$\therefore$ *0 ≤ median < 3*

$$\frac{\frac{m}{5} - 0}{} = 0.5$$

$$\frac{1}{5}m = 0.5$$

$m = 0.5 \times 5$

$m = 5 \times 0.5$

$m = 2.5$

$m = 2.5$ AG

A1

2

(e) $\quad \mathrm{E}(T) = \int_0^3 \frac{1}{5}t\,\mathrm{d}t + \int_3^6 \frac{1}{45}t^2(6-t)\,\mathrm{d}t$

M1

$$= \left[\frac{1}{10}t^2\right]_0^3 + \left[\frac{2}{45}t^3 - \frac{1}{180}t^4\right]_3^6$$

A1A1

$$= \frac{9}{10} + 1.65$$

$= 2.55$

A1

$\therefore$ P (median < T < mean)
= P (2.5 < T < 2.55)

M1

$$= 0.05 \times \frac{1}{5}$$

$= 0.01$

A1

6

[14]

Q23.

(a) $\quad k = 0.1$

OE.

B1

1

(b) E (X) = 1

B1

1

(c) P (X > 0) = 6 × 0.1

M1

= 0.6

A1

2

(d) P (| X | > 3.5) = 1 – P (| X | < 3.5)

M1

= 1 – 0.7

A1

= 0.3

A1

3

Alternative solution
P (X < –3.5) + P (X > 3.5)

(M1)

$= \frac{0.5}{10} + \frac{2.5}{10}$

(A1)

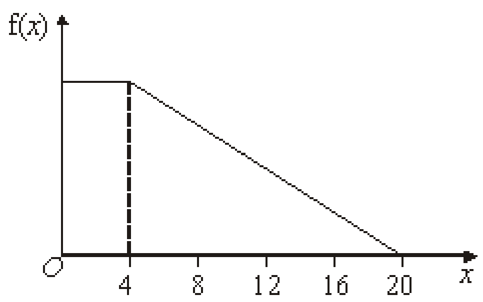
$= \frac{3}{10}$

(A1)

[7]

Q24.

(a)



horizontal axis; 0 to 20

B1

vertical axis; 0 to c or 1/12

B1

horizontal line @ c from 0 to 4

B1

Line from (4,c) to (20,0)

B1

4

(b) Area under graph = 1

use of

M1

Area under graph =

$4c + \frac{1}{2}(20 - 4)c = 12c$

area of (rectangle + triangle)

M1

or

$= \frac{c}{2}(4 + 20) = 12c$

area of trapezium

Hence 12 c = 1 so c = $\frac{1}{12}$

CAO; not decimal equivalent
(but accept 0.083)

A1

3

(c) P(Length < 2.01) = P (X < 10)

CAO

B1

f (10) = $\frac{5c}{8} = \frac{5}{96} = 0.0521$

CAO/AWRT; ft on c only

B1ft

P(X < 10) = 4c + $\frac{1}{2}\left(c + \frac{5c}{8}\right)6$

area of (rectangle + trapezium)

or $\int_2^4 c\,dx + \int_4^{10} \frac{c}{16}(20-x)\,dx$ *etc*

1 – area of (triangle)

M1

or

$= 1 - \frac{1}{2}(20-10)\frac{5c}{8}$

$= \frac{71c}{8}$ or $1 - \frac{25c}{8}$

[cx] $^4_0 + \left[\frac{c}{16}\left(20x - \frac{x^2}{2}\right)\right]_4^{10}$ *A1*

$= \frac{71}{96}$ or 0.739 to 0.740

CAO/AWRT; accept 0.74

A1

4

[11]

Q25.

(a) $P(2 \le X \le 3) = F(3) - F(2)$

M1

$= \frac{27}{64} - \frac{1}{8}$

$= \frac{19}{64} = 0.297$

CAO/AWRT

A1

2

(b) $f(x) = \begin{cases} \frac{3x^2}{64} & 0 \le x \le 4 \end{cases}$

M1A1

2

(c) (i) $E(X) = \int_0^4 x.f(x)\,dx$

$= \int_0^4 \frac{3x^3}{64}\,dx$

M1

$= \left.\frac{3x^4}{256}\right|_0^4$

A1ft

= 3

CAO

A1

3

(ii) Var (X) = $\int x^2 \mathrm{f}(x)\,\mathrm{d}x - [\mathrm{E}(X)]^2$

$= \int_0^4 \frac{3x^4}{64}\,dx - (3)^2$

M1

$= \left.\frac{3x^5}{320}\right|_0^4 - 9$

on their f(x) and μ^2

A1ft

$= 9\frac{3}{5} - 9$

$= \frac{3}{5}$ or 0.6

CAO

A1

3

[10]

Q26.

(a) Use of area for probability

may be implied in (a) or (b)

M1

(i) P(T < 10) = 0.9

CAO

A1

(ii) P(T > 5) = 0.6

CAO

A1

3

(b) P(T < 10 | T > 5) = $\frac{P(5 < T < 10)}{P(T > 5)}$

attempt at conditional probability

M1

correct form

A1

$= \frac{(i) - (1 - (ii))}{(ii)}$

m1

$= \frac{0.9 - (1 - 0.6)}{0.6} = \frac{0.5}{0.6}$

$= \frac{5}{6}$ Or 0.833

CAO/AWRT

A1

4

OR

(Conditional) area above 5 is 6a

attempt at area above 5

(M1)

(Conditional) area above 5 must = 1

area must = 1 used

(m1)

Thus $a = \frac{1}{6}$

(A1)

Thus area below 10 is (10 – 5)

$= \frac{5}{6}$ Or 0.833

CAO/AWRT

(A1)

Note:
many candidates will quote answers without any working so:

3 correct	*7 marks in total*
2 correct in (a)	*3 marks in (a)*
1 correct in (a)	*2 marks in (a)*
$\frac{x}{0.6} < 1$ *in (b)*	*2 marks in (b)*
$\frac{0.5}{0.6}$ *in (b)*	*3 marks in (b)*

[7]

Q27.

(a) (i) $X \sim R\ (a, a + k)$

V(X) = $\frac{(a+k+a)^2}{12} = \frac{k^2}{12}$

CAO
use of (b – a) must include = k

B1

= 48 $\Rightarrow$ $k^2 = 576$ $\Rightarrow$ $k = 24$

or $k = 24$ $\Rightarrow$ $\frac{k^2}{12} = \frac{24^2}{12}$ =48

CAO; AG

B1

2

(ii) E (X) = $\frac{a+k+a}{2} = a + \frac{k}{2}$

CAO; OE
use of (a + b) must include = (2a + k) OE

B1

6 => a + 12 = 6 => a = –6

or a = –6 => a + $\frac{k}{2}$ = –6 + 12 = 6

CAO; AG

B1

2

(b) P(X > 0) = $\frac{a+k-0}{a+k-a} = \frac{a+k}{k}$

attempt at area of a rectangle of height $\frac{1}{k}$

M1

$= \frac{3}{4}$ or 0.75

CAO

A1

2

[6]

Q28.

(a) $P(T = 0) = 0$

B1

1

(b) (i) $P(T < 2) = 1 - e^{\frac{-2}{4}} = 1 - e^{-0.5}$

M1

$= 0.393$

AWRT

A1

2

(ii) $P(2 \le T < 6) = F(6) - F(2)$

M1

$= e^{-0.5} - e^{-1.5}$

PI

m1

$= 0.383$

0.383 to 0.384

A1

3

(c) We require $P(T > 4 \mid T > 1)$

M1

$= \frac{P(T > 4 \text{ and } T > 1)}{P(T > 1)}$

m1

$$= \frac{e^{-1}}{e^{-0.25}}$$

m1

= 0.472

CAO

A1

4

[10]

Q29.

(a) P(T < 8) = (8 – 4) × $\left(\frac{0.05 \times 0.1}{2}\right)$

Trapezium

Worthwhile attempt at correct area, however divided

M1

or

{(8 – 4) × 0.05} + $\left\{\frac{1}{2} \times (8-4) \times (0.1-0.05)\right\}$

Rectangle + triangle

= 0.3

CAO; OE

A1

2

(b) (i) Area under graph = 1

Use of; may be implied by their area;

accept P(*T > 8) = 1 – (a) must be stated clearly in reverse method*

M1

Area = (a) + {(s – 8) × 0.1} + $\left\{\frac{1}{2} \times (20-s) \times 0.1\right\}$

Worthwhile attempt at area under given graph or area above 8, however divided

M1

or

$$\left(\frac{(20-8)+(s-8)}{2}\right) \times 0.1$$

Hence $0.05s = 0.5$
(implies $s = 10$)

CAO; OE

AG

NB: In reverse method, assuming
s = 10 so triangle area = 0.5 then showing
s = 10 given rectangle area = 0.2, scores max of M1M1A0

A1

3

(ii) $P(T > 15) = \frac{1}{2} \times (20 - 15) \times f(15)$

Area of correct triangle or $\int_{15}^{20} y\,dx$

M1

However using (b)
$f(15) = 0.05$

CAO; OE or $y = 0.2 - 0.01x$

B1

Thus $P(T > 15) = 0.125$

CAO; OE

A1

3

[8]

Q30.

(a) Mean, $\mu = 21 = \frac{a+b}{2}$

CAO; stated or used

B1

Variance, $\sigma^2 = 27 = \frac{(b-a)^2}{12}$

CAO; stated or used

B1

so
$((42 - a) - a)^2 = 12 \times 27 = 324$
or $b - a = (\pm)\ 18$

Substitution of μ into σ^2 or $\sqrt{\ }$ of equation involving σ^2

M1

Thus (42 – 2a) = (±) 18

Solving quadratic or two simultaneous equations

M1

or $a + b = 42$ and $b - a = (\pm)\ 18$

Thus a = 30 or 12 and b = 12 or 30

As $a < b$ so a = 12 and b = 30

CAO; must state a < b

B1 for (12, 30) $\Rightarrow$ *μ = 21*

B1 for (12, 30) $\Rightarrow$ *σ^2 = 27*

A1

5

(b) (i) P(5 < X < 20) = P(12 < X < 20) =

Lower limit of 12 or 20 to 30

B1

$\frac{20-l}{b-a}$ or $1 - \frac{30-20}{b-a}$

Attempt at area of a rectangle of height $\frac{1}{b-a}$ *or* $\frac{1}{18}$

Can be scored in (ii)

M1

= 8/18 or 4/9 or 0.44

CAO/AWRT; OE

A1

3

(ii) $P\left(X < \mu - \frac{\sigma\sqrt{3}}{2}\right) =$

$P\left(X < 21 - \frac{\sqrt{27}\sqrt{3}}{2}\right) =$

Substitution of μ = 21 and σ = $\sqrt{27}$ *; OE*

M1

P(X < 16.5)

CAO

A1

= 4.5/18 or 1/4 or 0.25

CAO; OE

A1

3

[11]

Q31.

(a) f(6) = $\frac{2}{5}$ = 0.4

P($T \geq 6$) = $\frac{1}{2} \times \frac{3}{2} \times \frac{2}{5}$

M1

= $\frac{3}{10}$ or 0.3

A1

2

(b) $\int_3^m \frac{1}{90} t^2 \mathrm{d}t$ = 0.5

Correct limits

M1

$\int f(t)\mathrm{d}t$ *= 0.5*

M1

$\therefore \left(\frac{1}{270} t^3\right)_3^m$ = 0.5

$\int \frac{1}{90} t^2 \mathrm{d}t$

m1

$= \frac{1}{270} t^3$

A1

$$\left.\begin{aligned} m^3 - 27 &= 135 \\ m^3 &= 162 \end{aligned}\right\}$$

Substitution of correct limits to obtain a cubic

m1

$m = 5.45$

CAO

A1f.t.

6

(c) $$\int_3^6 \frac{1}{90}t^3\,dt + \int_6^{7.5}\left(2t - \frac{4}{15}t^2\right)dt$$

Attempt at:

M1

(i) tf(t)

m1

(ii) two integrals

$$\left[\frac{t^4}{360}\right]_3^6 + \left[t^2 - \frac{4}{45}t^3\right]_6^{7.5}$$

(iii) correct integration

A1A1

(3.6 – 0.225) + (18.75 – 16.8)
3.375 – 1.95
= 5.325

$5\frac{13}{40}$ *(AWRT 5.33)*

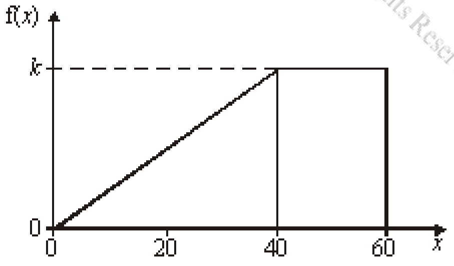
A1

5

[13]

Q32.

(a)



x-axis; (0) to60

B1

f(x)-axis; (0) to k or 0.025

B1

+ve slope straight line; 0 to 40

B1

horizontal straight line; 40 to 60

*(allow **minor** extensions)*
(0 for axes reversed)

B1

4

(b) Area under graph = 1

*use of; may be implied by their area **must** be stated for k = 0.025 assumed*

M1

Area = $\left(\frac{1}{2}\times 40\times k\right)+(20\times k)$

area of (triangle + rectangle)
[= 0.5 + 0.5 (+A1)]

M1

or

Area = $k\times\left(\frac{60+20}{2}\right)$

area of (trapezium)
[= 1 (+ A1)

= 40k

cao; or equivalent

A1

3

(implies k = 0.025)

AG
(Area = 40k $\Rightarrow$ M0 M1 A1)

(c) At x = 30 height = 0.75k or 0.0188

cao/awrt or equivalent

B1

or $\left[\frac{kx^2}{80}\right]_0^{30}$

or equivalent

P(X > 30) = $\left(10\times\left(\frac{0.75k+k}{2}\right)\right)$ + (20 × k)

area of (trapezium + rectangle)

M1

or $1-\left(\frac{1}{2}\times 30\times 0.75k\right)$

1 – area of (triangle)

= 28.75k or (1 – 11.25k)

115k/4 or (1 – 45k/4)

=23/32 or 0.719

cao/awrt (0.71875

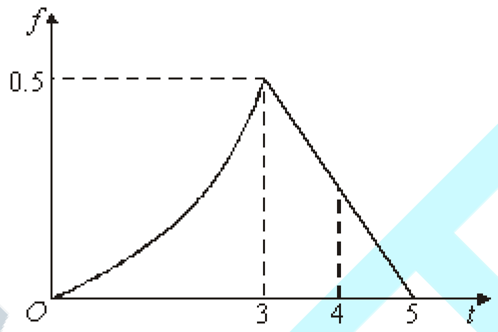
A1

3

[10]

Q33.

(a)



Curve from(0, 0) to (3, 0.5)

B1

Straight line from (3, 0.5) to (5, 0)

B1

2

(b)

F(t) =

$$\begin{cases} \frac{t^3}{54} & 0 \le t \le 3 \\ \frac{1}{8}(10t = t^2 - 17) & 3 \le t \le 5 \end{cases}$$

B1

M1M1

A1

4

(c) P(T < 4) = F(4) = $\frac{1}{8}$ (40 – 16 – 17)

M1

$\frac{7}{8}$ or 0.875

Alternative (c): $1 - \frac{1}{2} \times 1 \times f(4)$

$1 - \frac{1}{2} \times 1 \times \frac{1}{4}$

$1 - \frac{1}{8} = \frac{7}{8}$

A1

2

[8]

Q34.

(a) Area = 1

Use of

M1

Area = $\frac{4 \times 2c}{2} + (2 \times 2c) + \frac{12 \times 2c}{2}$

Attempt at area of (2 triangles + rectangle) or equivalent

M1

= 20*c*

CAO

A1

3

$\therefore$ 20*c* = l $\Rightarrow$ *c* = 0.05

AG

(b) (i) P(*X* > 4) = P(4 < *X* < 6) + P(*X* > 6)

M1

or = 1 – P(*X* < 4)

Attempt at area of (rectangle + triangle) or (1 – triangle) or equivalent

= 4*c* + 12*c* = 1 – 4*c* = 0.8

CAO; or equivalent

A1

2

(ii) P(4 < *X* < 12) = P(*X* < 12) – P(*X* < 4)

use of; or equivalent

M1

= {(1 – 3*c*) or (4*c* + 4*c* + 9*c*)} – (4*c*)

= (1 – 7c) or 13c

either

A1

= 1 – 0.35 = 0.65

CAO; or equivalent

A1

3

(iii) P(X < 12|X > 4) = $\frac{P(4 < X < 12)}{P(X > 4)}$)

Attempt at conditional probability

M1

Correct expression

A1

$= \frac{\text{(ii)}}{\text{(i)}} = \frac{0.65}{0.80}$

m1

$= \frac{13}{16} = 0.8125$

CAO or AWFW 0.812 to 0.813

A1

4

NB *Area > 4 is 16c so for conditional*

(M1)

distribution, c = 0.0625

(A1)

Area < 12 is 13c for this distribution

(m1)

Thus probability = 13 × 0.0625 = 0.8125

(A1)

(c) Some delays greater than 18 minutes
Some appointments early
PDF unlikely to be linear

E1

1

[13]

Q35.

(a) (i) Mean = μ = 4c

CAO

B1

Variance = $\sigma^2 = 3c^2$

CAO

B1

2

(ii) $E(X^2) = Var(X) + (E(X))^2$

use of or equivalent

M1

$= 3c^2 + (4c)^2$

A1ft

2

$= 19c^2$

AG

(b) $19c^2 = 171$

$\therefore c = 3$

CAO

B1

1

(c) (i) $P\left(X > \frac{\mu}{2} + \frac{\sigma}{\sqrt{3}}\right) = P(X > 6 + 3)$

$= P(X > 9)$

CAO

B1

$= \frac{7c - 9}{6c}$ or $1 - \frac{9 - c}{6c}$

attempt at correct area

M1

$= \frac{2}{3} = 0.67$

CAO/AWRT

A1

3

(ii) $P(X < d) = 0.25$

$P(X < d) = \frac{d - c}{6c} = \frac{d - 3}{18}$

*attempt at correct area **and** substitution of their value of c*

M1

$\therefore \frac{d - 3}{18} = 0.25$

Equating their expression in d to 0.25

m1

$\therefore$ d = 7.5

CAO

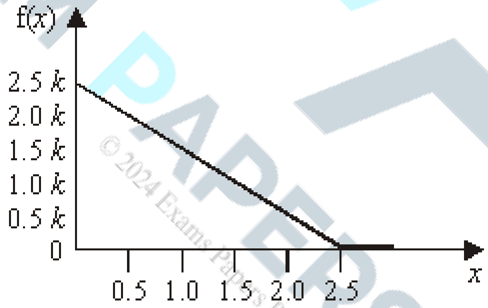
A1

3

[11]

Q36.

(a)



x-axis: 0 to 2.5

B1

f(x)-axis: 0 to 2.5k (or 0.8);
assume starts at 0 if not stated

B1

straight line with negative slope;
ignore minor extensions;
(0, 2.5k or 0.8) to (2.5, 0)

B1

3

(b) Area under line = 1

use of (may be implied)

M1

Area = $\frac{1}{2} \times 2.5 \times 2.5k$

Attempt at area (0 to 2.5)

M1

Or = $\int_0^{2.5} k(2.5 - x)\,dx$

attempt at integration (0 to 2.5)

= $3.125k$

(= 1 implies that $k = 0.32$)

CAO or equivalent

AG

NB accept reverse argument

A1

3

(c) P($1 < X < 2$):

f(1) = $1.5k = 0.48$

f(2) = $0.5k = 0.16$

CAO both may be implied

B1

Area = $(2 - 1) \times \frac{(0.48 + 0.16)}{2}$

Attempt at area of trapezium (1 to 2) or equivalent

M0 for 0.48 – 0.16 = 0.32

M1

0.32 = 8/25

CAO

A1

or

Area = $\int_1^2 0.32(2.5 - x)\,dx$

Attempt at area (1 to 2)

(M1)

$\left[0.8x - 0.16x^2\right]_1^2 =$

Fully correct expression; allow switch of limits

(A1)

(1.6 – 0.64) – (0.8 – 0.16) = 0.32 = 8/25

CAO

(A1) 3

[9]

Q37.

(a) f(x) = 1/4c $\quad -c < x < 3c$

Mean, $\mu = \frac{-c+3c}{2} = c$

CAO

B1

Variance, $\sigma^2 = \frac{(3c-(-c))^2}{12} = \frac{16c^2}{12} = \frac{4c^2}{3}$

CAO

B1 2

(b) E(X^2) = Var(X) + (E(X))^2 = $\sigma^2 + \mu^2$

∴ *use of, or* $\int x^2$ *f(x)*dx

M1

$\therefore \frac{28}{3} = \frac{4c^2}{3} + c^2 = \frac{7c^2}{3}$

substitution of answersin (a) or in intergral plus = $\frac{28}{3}$

m1

∴ 7c^2 = 28 or 3c^2 = 12 $\Rightarrow$ c = 2

CAO; AG
NB accept reverse argument

A1 3

(c) (i) P(X = 2) = zero

B1 1

(ii) P(|X| < 2.5) = P(–2.5 < X < 2.5)

= P(–2 < X < 2.5)

CAO; seen or implied

B1

$$\frac{2.5-(-2)}{8} = 1-\left(\frac{6-2.5}{8}\right)$$

attempt at area of a rectangle or attempt at integration

M1

$= \frac{9}{16}$ = 0.562 to 0.563

CAO / AWFW

A1

3

[9]

Q38.

(a) P(T = 4) = 0

B1

1

(b) median = 3 $\Rightarrow$ P($T \le 3$) = 0.5

P(T > 3) = $\frac{1}{2}$ × f(3) × (5 – 3)

M1

= $\frac{1}{2}$ × 0.5 × 2

= 0.5

A1

2

(c) P($T \ge 3.5$) = $\frac{1}{2}$ × f(3.5) × (5 – 3.5)

M1

= $\frac{1}{2}$ × $\frac{3}{8}$ × 1.5

M1A1

= $\frac{9}{32}$

0.28125

A1

$\therefore$ P(T < 3.5) = 1 – $\frac{9}{32}$

= $\frac{23}{32}$

0.71875 (AWRT 0.719)

A1

5

(d) E(T) = $\int_0^3 \frac{t^3}{18}\,dt + \frac{1}{4}\int_3^5 t(5-t)\,dt$

M1

= $\left[\frac{t^4}{72}\right]_0^3 + \left[\frac{1}{4}\left\{\frac{5t^2}{2} - \frac{t^3}{3}\right\}\right]_3^5$

Integration attempted

M1A1

= $\frac{9}{8} + \frac{1}{4}\left\{\left(\frac{125}{2} - \frac{125}{3}\right) - \left(\frac{45}{2} - 9\right)\right\}$

Substituting limits

M1

= $1\frac{1}{8} + 1\frac{5}{6}$

= $2\frac{23}{24}$

2.9583 (AWRT 2.96)

A1

5

[13]

Q39.

(a) Area under graph = 1

use of; may be implied by their area
Must be clearly stated in reverse argument

M1

Area = 20c + 2 × 20 × $\left(\frac{c+3c}{2}\right)$ = 20c + 80c

or $= 60c + 2 \times \left(\frac{1}{2} \times 20 \times 2c\right) = 60c + 40c$

worthwhile attempt at area under given graph, however divided

statement of form 100c = 1 $\Rightarrow$ M1M0A0

Use of heights $\Rightarrow$ M0M0A0

M1

$= 100c$

(= 1 implies $c = 0.01$)

CAO; stated or clear

AG

NB accept reverse argument

A1

3

(b) (i) $P(X > 20) = 10 \times \left(\frac{2c + 3c}{2}\right) = 25c$

or $= 10c + \left(\frac{1}{2} \times 10 \times (3c - 2c)\right) = 10c + 5c$

identification of 2c or 0.02 and worthwhile attempt at correct area, however divided

M1

$= 25c = \frac{1}{4}$ or 0.25

CAO; OE

A1

2

(ii) $P(X < 10) = 20c + 20 \times \left(\frac{c + 3c}{2}\right) = 20c + 40c$

or $= 40c + \left(\frac{1}{2} \times 20 \times 2c\right) = 40c + 20c$

M1

worthwhile attempt at correct area, however divided

or $1 - 20 \times \left(\frac{c + 3c}{2}\right) = 1 - 40c$

or $= 1 - 20c - \left(\frac{1}{2} \times 20 \times 2c\right) = 1 - 20c - 20c$

$= 60c = \frac{3}{5}$ or 0.6

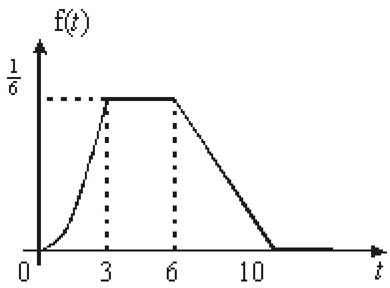
CAO; OE

A1 2

[7]

Q40.

(a)



1 for each correct part of graph

B4 4

1 for axes $\left(\frac{1}{6}\ and\ 10; f(t); t\right)$

(b) (i) $P(3 \le T < 6)$ = Area of rectangle

M1

$= 3 \times \frac{1}{6}$

$= \frac{1}{2}$

A1 2

(ii) $P(T > 6)$ = Area of triangle

M1

$= \frac{1}{2} \times \frac{1}{6} \times 4$

$= \frac{1}{3}$

(accept $0.3\dot{3}$ *)*

A1 2

(iii) $P(T < 3) = 1 - \left(\frac{1}{3} + \frac{1}{2}\right) = \frac{1}{6}$

$\therefore$ *(seen anywhere) or* $\int_0^3 f(t)\,dt = \frac{1}{6}$

B1

$P(T <$ lower quartile$) = \frac{1}{4}$

$P(T > Q_1) = \frac{3}{4}$ or $\int_0^{Q_1} f(t)\,dt = \frac{1}{4}$

M1

$\therefore$ lower quartile lies in the range $3 < T < 6$

$\therefore$ lower quartile $= 3 + d$

Area $= d \times \frac{1}{6} = \frac{1}{4} \times \frac{1}{6} = \frac{1}{12}$

Any valid attempt at d

M1

where $d = \frac{1}{12} \div \frac{1}{6} = \frac{1}{2}$

$\therefore$ lower quartile $= 3 + \frac{1}{2} = 3.5$

CAO

A1 **4**

(c) $P(T < 2) = \int_0^2 \left(\frac{1}{54}t^2\right) dt$

(ignore limits)

M1

$= \left\{\frac{1}{165}t^3\right\}_0^2$

(for $\frac{1}{162}t^3; OE$ *)*

A1

$= \frac{4}{81}$

$= 0.0494$ (3sf)

= 5%

$$\left.\begin{array}{l} = \frac{4}{8} \\ = 0.0494\ (3sf) \\ \approx 5\% \end{array}\right\}$$

(Substitution of correct limits)

m1

any correct form (AWRT 0.049)

A1

4

[16]

Q41.

(a) (i) $f(t) = 2e^{-2t}$, $(t \geq 0)$

no penalty for omitting $t \geq 0$

B1

1

(ii) Mean = 0.5

CAO

B1

Standard deviation = 0.5

CAO

B1

2

(b) $P(T > 3) = 1 - F(3)$

$= 1 - (1 - e^{-6})$

or by integration

M1

= 0.0248

AWRT 0.00248

A1

2

(c) Median m satisfies $F(m) = 0.5$

$1 - e^{-2m} = 0.5$

or by integration

M1

$e^{-2m} = 0.5$

$-2m = \ln (0.5)$

or $e^{2m} = 2$

$m = -0.5 \ln(0.5)$ A1

giving 0.5 ln (2)

(= 0.347)

Median delay = 20.82
= 21 seconds to nearest second

CAO

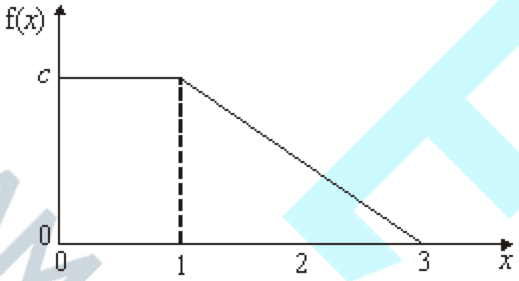
A1

3

[8]

Q42.

(a)



x-axis:0 to 3

B1

f(x)-axis; 0 to c

B1

horizontal line: 0 to 1 at c

B1

negative slope line;

B1

4

(1, c f.t.) to (3, 0)

(b) Area under graph = 1

Use of

May be implied by their area

M1

Area = $c + \frac{2 \times c}{2}$ or $c \times \left(\frac{3+1}{2}\right)$

Area of (rectangle + triangle)

or

Area of (trapezium)

M1

$= 2c$

Thus $2c = 1$ implies $c = \frac{1}{2}$ or .0.5

CAO

A1

3

(c) $f(2) = c/2 = \frac{1}{4}$ or 0.25

ft on c; may be implied

B1f.t.

$P(X < 2) = 1 - P(X > 2)$

$= 1 - \frac{1 \times (c/2)}{2} = 1 - \frac{c}{4} = 1 - \frac{1}{8}$

1 – area of triangle

Or

M1

$= c + 1 \times \left(\frac{c + (c/2)}{2}\right) = \frac{7c}{4} = \frac{7}{8}$

Area of (rectangle + trapezium)

$= \frac{7}{8}$ or 0.875

CAO

A1

3

[10]

Q43.

(a) Mean $\mu = \frac{a + (a + k)}{2} = \frac{2a + k}{2}$ or $a + \frac{k}{2}$

B1

Variance $\sigma^2 = \frac{((a + k) - a)^2}{12} = \frac{k^2}{12}$

CAO

B1

2

(b) Attempt at using results from (a)

M1

Var(X) = 3 $\Rightarrow$ $\frac{k^2}{12}$ = 3 or k^2 = 36

or equivalent; must be stated

AG

A1

$\therefore$ k = 6

E(X) = 21 $\Rightarrow$ a + 3 = 21

$\therefore$ a = 18

or equivalent; must be stated

AG

[Accept reverse arguments]

A1

3

(c) P(20 < X < 25) = P(20 < X < 24)

Upper limit of 24 or (a + k)

B1

$= \frac{24 - 20}{k}$

Area of rectangle of height k

M1

$= \frac{2}{3}$ or 0.67

CAO/AWRT

A1

3

[8]

Q44.

In parts (a) and (c), accept answers based on counting squares and triangles

(a) Area = 1

Use of

M1

Area = $\frac{2(4c+6c)}{2} + \frac{3(6c+3c)}{2}$

Attempt at area
≥ 1 correct area even if split into 2 rectangles and 2 triangles

M1/A1

=10c + $\frac{27c}{2} = \frac{47c}{2}$

Area = 23.5c or equivalent

A1

4

∴ $c = \frac{2}{47}$

AG

(b) Zero

CAO

B1

1

(c) P(249 < X < 252) =

Correct area(s) identified and attempted
Accept split into triangles and rectangles.

M1

Either:
P(249 < X < 250) + P(250 < X < 252) =

$\frac{(5c+6c)}{2}+\frac{2(6c+4c)}{2}=$

$55c + 10c = 15.5c$

or:

$1-\frac{(4c+5c)}{2}-\frac{(4c+3c)}{2}=$

$1 – 4.5c – 3.5c = 1 – 8c$

Correct expression

A1

$=\frac{31}{47}=0.66$

CAO/AWRT

A1

3

[8]

Q45.

(a) $\mu = 201$

CAO (ignore how obtained)

B1

$\sigma^2 = 3$

cao

B1

2

(b) $f(x) = \frac{1}{6}$

CAO; accept AWRT 0.167 (accept $h = k = \frac{1}{6}$ written or on diagram unless followed by f(x) = 6)

B1

1

(c) (i) $P(X = \mu) = 0$

CAO

B1

1

(ii) $P(X < 200 – \sigma) =$

$$\frac{(200-\sigma)-198}{6} = 1 - \frac{(204-(200-\sigma))}{6}$$

use of area of rectangle or integral (allow for 200 – σ or 200 – σ² < 198 providing also ∴ prob = 0)

M1

$$= \frac{2-\sqrt{3}}{6} = 1 - \frac{(4+\sqrt{3})}{6}$$

must use their $\sqrt{\sigma^2}$ *and their f(x) in formula or integral*

A1f.t.

$(2 - \sqrt{3}) / 6 = 0.045$

CAO OE / AWRT

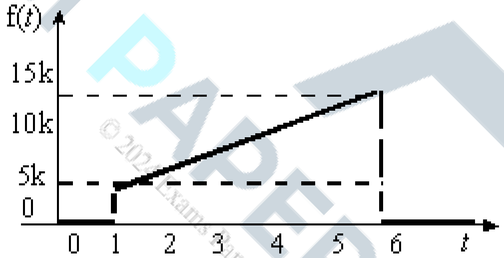
A1

3

[7]

Q46.

(a)



t-axis; 1 – 6(can start at 1)

B1

f(t) axis; 0 to 15k or 0 to 0.3

B1

(assume 0 if not stated)

B1

3

straight line;
(1,5k) to (6,15k) or (1,0.1) to (6,0.3);
ignore minor extensions

(b) Area = 1

use of (may be implied)

M1

Area = $5 \times \left(\frac{5k+15k}{2}\right)$

attempt at area of trapezium or rectangle + triangle

or $\int_1^6 k(2t + 3)dt$

M1

$= (5 \times 5k) + \left(\frac{1 \times 5 \times 10k}{2}\right)$

$= 50k$

CAO;AG

A1

3

$(= 1 \Rightarrow k = 1/50)$

(c) (i) f(3) = 9k or 0.18

CAO

B1

$P(T < 3) = (3 - 1)\left(\frac{5k + 9k}{2}\right)$

attempt at correct area, howeverdivided or attempted integration (1,3) with A1 for k(t² + 3t) and limits (1,3)

M1

$= (2 \times 5k) + \left(\frac{1 \times 2 \times 4k}{2}\right)$

$= 14k = 0.28$

CAO OE

A1

3

(ii) f(2) = 7k or 0.14

CAO both

B1

f(5) = 13k or 0.26

$P(2<T<5)$

$= (5-2)\left(\frac{7k + 13k}{2}\right)$

attempt at correct area, however divided or attempted integration (2,5) with A1 for k(t² + 3t) and limits (2,5)

M1

$30k = 0.6$

CAO OE

NB 1 – 0.14 – 0.26 = 0.6 $\Rightarrow$ *B1M0A0*

A1

3

[12]

Q47.

(a) $\int_1^5 ks^2$ ds = 1

for $\int ks^2 \mathrm{d}s$

M1

$\Rightarrow \left[\frac{ks^3}{3}\right]_1^5 = 1$

for $\frac{ks^3}{3}$

A1

$\Rightarrow \frac{124}{3}k = 1$

$k = \frac{3}{124}$

AG

A1

4

(b) P(S < 3) = $\frac{3}{124}\int_1^3 s^2 \mathrm{d}s$

1 – $\int_3^5 ks^2 \,\mathrm{d}s$

M1

$= \left[\frac{s^3}{124}\right]_1^3$

(limits must be correct for M1)

$= \frac{27}{124} - \frac{1}{124}$

$= \dfrac{26}{124}$

substitute limits $1 - \left(\dfrac{125}{3} - \dfrac{27}{3}\right)k$

$= 1 - \dfrac{49}{62} \quad = \dfrac{13}{62}$

m1

$= \dfrac{13}{62}$

AWRT 0.210

A1

3

(c) $\dfrac{3}{124}\int_1^m s^2\,ds = 0.5$

$k\int_m^5 s^2\,ds = 0.5$

M1

$\left[\dfrac{s^3}{124}\right]_1^m = 0.5$

$\left[\dfrac{s^3}{124}\right]_m^5 = 0.5$

A1

$m^3 - 1 = 62$

$125 - m^3 = 62$

M1

$m^3 = 63$

$m = \sqrt[3]{63}$

$m = 3.98$ (3SF)

$m = 4$ min

A1

4

[11]

Q48.

(a) So that area = (512 – 500) × $\dfrac{1}{12}$ = 1

Area = 1 as a result

E1

or $\frac{1}{12} = \frac{1}{k - 500}$

k = 512 as a result

1

(b) Mean, $\mu = \frac{512 + 500}{2} = 506$

CAO

B1

Variance, $\sigma^2 = \frac{(512 - 500)^2}{12} = 12$

Use of; allow even if called standard deviation

M1

Thus standard deviation, $\sigma = \sqrt{12} = 3.46$

CAO/AWRT

A1

3

(c) (i) $P(X < 509) = \frac{509 - 500}{12}$

Attempt at area of rectangle below 509 or of integral from 500 to 509

M1

$\frac{3}{4} = 0.75$

CAO

M1

2

(ii) $P(X > \mu - \sigma) = \frac{512 - (\mu - \sigma)}{12}$

Attempt at area of rectangle above ($\mu - \sigma$) or of integral from ($\mu - \sigma$) to 512
allow use of value of σ^2 only if subsequent answer is stated as 1

M1

$\frac{512 - (506 - \sqrt{12})}{12}$

$\sqrt{}$ on μ and σ in formula or integral; not on σ^2

A1f.t.

$\dfrac{6+\sqrt{12}}{12}$ = 0.788 to 0.789

AWFW

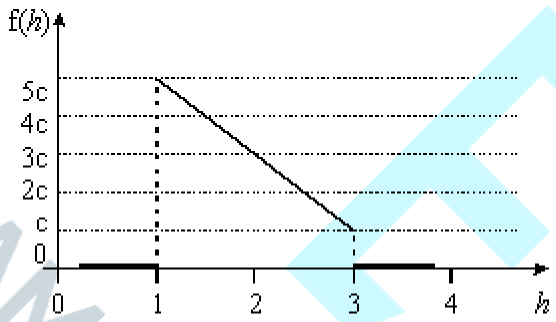
A1

3

[9]

Q49.

(a)



h-axis: 1 to 3; can start at 1

B1

f(h)-axis: 0 to 5c or 0 to $\frac{5}{6}$;

B1

assume starts at 0 if not stated

B1

Straight line, negative slope which must not meet h axis;

ignore minor extensions (1,5c) or $\left(1, \frac{5}{6}\right)$ *& (3,c) or* $\left(3, \frac{5}{6}\right)$

3

(b) Area under line = 1

Use of (may be implied)

M1

Area = $(3-1)\times\dfrac{5c+c}{2}$

Attempted at area of trapezium

= $[(3-1)\times c]+\left[\dfrac{1}{2}\times(3-1)\times(5c-c)\right]$

Attempt at area of rectangle plus triangle

M1

$= \int_1^3 c(7-2h)\,dh$

Attempt at integration

= 6*c*

CAO

A1

$\left(= 1 \text{ implies } c = \frac{1}{6}\right)$

AG

3

(c) Area = $(3-2)\times\frac{(3c+c)}{2}$

$= [(3-2)\times c]+\left[\frac{1}{2}\times(3-2)\times(3c-c)\right]$

Attempt at correct area

M1

$= \int_2^3 c(7-2h)\,dh$

$= 2c = \frac{1}{3}$

CAO; accept AWRT 0.333

A1

2

(d) $P\left(2 < H < 2\frac{1}{2}\right) =$

Attempt at

M1

$\frac{1}{2}\times\left(\frac{3c+2c}{2}\right)=\frac{5c}{4}=\frac{5}{24}=0.208$

CAO/AWRT

A1

$P\left(H < 2\frac{1}{2} \middle| H > 2\right) = \frac{P\left(2 < H < 2\frac{1}{2}\right)}{P(H > 2)}$

Attempt at conditional probability

M1

$= \frac{\frac{5}{24}}{\text{part (c)}}$

m1

$\frac{5}{8} = 0.625$

CAO

A1

5

[13]

Q50.

(a) (i) 1

1 cao

B1

(ii) 1.54

1.54 cao

B1

(iii) 1.56

1.56 cao

B1

(iv) 1.73

1.73 cao

B1

(v) 1

1 cao

B1

5

(b) (i) 0.5 – 0.291 = 0.209

method

M1

0.209 cao

A1

2

(ii) $\sqrt{0.0751} = 0.2474$

method

M1

0.274 (0.2735 to 0.2745)

A1

2

allow B1 for variance = 0.0751

(iii) $E(X^2) - 1.54^2$ -= 0.0751

any correct equation

M1

$E(X^2) = 2.45$

2.45 (2.44 to 2.45)

A1

2

[11]

Q51.

(a) Maximum of p.d.f.

Maximum of graph

E1

1

(b) (i) Variance = $0.369 - 0.542^2 = 0.075236$

Method for variance

M1

s.d. = $\sqrt{0.075236} = 0.274$

Method for s.d.- disallow if called variance

m1

0.274 (0.274 – 0.275)

***SC** allow A1 for variance = 0.0752 (0.0752 – 0.0753)*

A1

3

(ii) 1 – 0.719 = 0.281

Use of 0.719 *0.281 cao*

B1B1

2

(iii) 0.557

0.557 cao

B1

1

(iv) 0.5 – 0.483 = 0.017

Use of 0.483, allow 1 – 0.483

B1

Correct method

M1

0.017 cao

A1

3

[10]

Q52.

(a) $\int_{-6}^{14} k \, dy = [ky]_{-6}^{14}$ = k(14 – (–6)) = 20k

Allow by diagram – generous

M1

20k = 1 k = 0.05

0.05 ***ag***

A1

2

(b) 4

4 cao

B1

1

(c) (i) $\int_{-6}^{14} 0.05y^2 \, dy = \left[\frac{0.05y^3}{3}\right]_{-6}^{14}$

Any correct expression for E(Y^2) – ignore limits

Correct method – arithmetic/sign errors only

M1 m1

= 45.733 – (–3.6) = 49.3

49.333 (49.3 – 49.35)

A1

3

(ii) Var(Y) = 49.3333 – 4^2 = 33.3333

Method for Var – their E(Y^2)

M1

s.d. = $\sqrt{33.3333}$ = 5.77

Requires all previous method marks in (c)
5.77(5.77 – 5.775)

m1 A1 3

(d) (i) 0.3
0.3 acf

B1 1

(ii) $\int_{-2}^{2} 0.05 dy = [0.05y]_{-2}^{2}$
Allow diagram

M1

= 0.2
0.2 ***ag***

A1 2

(ii) 0.6
Allow diagram
0.6 acf

M1 A1 2

[14]

Q53.

(a) (i) c/2
cao

B1

(ii) $\int_0^c x^2 / c \, dx = [x^3 / 3c]_0^c = c^2 / 3$

any correct expression - allow no limits
correct evaluation of a definite integral

M1 M1

variance = $c^2/3 - (c/2)^2 = c^2/12$
for variance - their E(X) and E(X^2)
requires all previous M marks
(answer given)

M1 A1 5

(b) $\int_{c/4}^{c} \frac{1}{x} \mathrm{d}x = \left[\frac{x}{c}\right]_{c/4}^{c}$

method - integration or diagram
¾ cao acf

M1 A1

2

(c) (i) 14 is estimate of *c*/2
any correct expression involving estimate of c

M1

28 is estimate of *c*

A1

28/√12 = 8.08 is estimate of standard

m1

deviation of X
for standard deviation
8.08 ~ 8.1

A1

(ii) 1028 grams
or 1030

B1ft

5

[12]

Examiner reports

Q2.

Students should not have expected to gain full marks when they failed to use a ruler to help them draw the straight lines required to complete the graph. *Freehand sketches* were not good enough since they were asked to *draw* a graph. Part (b) was usually completed successfully. Although students produced some excellent solutions to part (c), these were not always fully explained. Only the very best students managed a fully correct solution to all parts of (c). Many, in part (c)(iii), simply multiplied together their answers to parts (c)(i) and (c)(ii), obviously believing that the events $X > 2.5$ and $1.5 < X < 4.5$ were independent. This was incorrect and so gained no credit. Many students used calculus to find the required areas, often with more success than those who attempted to use a combination of areas of simple geometric shapes (rectangle, trapezium, triangle). Students using the latter method seemed unable, on the whole, to cope with calculations involving fractions, often writing $\frac{5}{12}$ as 0.41 or 0.416 before attempting to calculate the required area. Marks were often lost, especially in parts (c)(iii) and (c)(iv), due to students' inability to write their answers in an **exact** form as requested. The correct answer to part (c)(iii) of $\frac{17}{48}$ was often seen as the answer to part (c)(iv). Some students gave excellent solutions to this question by first finding the cumulative distribution function F(x). Unfortunately, some weaker students who attempted this method were unable to find the correct form for F(x) and so lost marks.

Q3.

The graph in part (a) was usually adequately sketched. There was, however, a minority of candidates who interpreted the request for a 'sketch' as an excuse to offer up a very poor quality freehand drawing. This should not have been the case and any such attempts were likely to be penalised.

The majority of candidates used simple areas of rectangles to obtain their very efficient solutions to parts (b)(i) and (b)(ii). Most candidates realised that, when the answers are given in the question, sufficient working must be shown in order to obtain full credit. Some candidates, when they are confronted by a continuous distribution, seem to think that integration methods always have to be used. This is not the case when straight lines form part of the graph of the probability density function. Those candidates who used valid integration methods usually gained the correct given values and thus full credit.

Candidates were expected to use the information given in part (b) to write down the answers to parts (c)(i) and (c)(ii).

Although some candidates recognised that the answers to part (b) implied that the upper quartile of X was $8\frac{1}{3}$ 8 3 and that the lower quartile of X was 3, many attempted to recalculate these values, usually incorrectly, and mostly by the use of calculus methods.

Disappointingly, in part (d), only the most able candidates recognised that, whatever the values obtained for the median and lower quartile, P (X < median | $X \geq$ lower quartile) = $\frac{1}{3}$ and P (X < median **and** | $X \geq$ lower quartile) = $\frac{1}{4}$ and are always true.

Q4.

Although most candidates managed to correctly integrate $\frac{3}{8}(x^2+1)$ in part (a), many then failed to use the correct limits required to find F(x), with the vast majority failing to consider any limits at all but simply stating the given answer.

In part (b), many candidates stated F (m) = 0.5 but then failed to verify that m = 1 satisfied this condition. Part (c) was not well answered except by the more able candidates. Many started from the incorrect assumption that $\int_0^q \frac{1}{4}(5-2x)\mathrm{d}x = \frac{3}{4}$. Consequently, the given equation was often quoted without any foundation at the end of a lot of spurious work.

It was also very disappointing to see candidates on this paper unable to solve the given quadratic equation. Those who did manage to obtain two surd answers were often then unable to give an adequate numerical comparison in order to justify the selection of the given answer. Most candidates simply wrote down the given answer without any justification whatsoever. This lost them the final two marks for this part of the question. Part (d), although relatively straightforward, confounded many candidates as very few realised that all they had to do was to evaluate
F (1.5) – F(q) where F(q) = ¾ and F(1.5) = 13/16.

Q5.

Surprisingly, this was the worst attempted question on the paper. The majority of candidates gained both marks in part (a), but many failed to recognise that the continuous random variable, T, followed a rectangular distribution. Consequently, there were many attempts using calculus to find the values and, although these were often correct, this wasted time that could perhaps have been better used in answering other more difficult questions.

Part (b) was only answered correctly by the better candidates. Many did not understand that P($-2 < T < 2$) was required and so simply attempted to evaluate P($T > 2$) with the result that 23/30 was the most prevalent incorrect answer. Too many candidates treated T as a discrete random variable with P($T > 2$) = 1 – P($T \le 1$) often seen, whilst others again used unnecessary integration rather than simple areas of rectangles.

Q6.

In the previous Report on the Examination, it stated that too many candidates seemed to be under the misapprehension that a request for a ‘sketch’ entitled them to draw a very poor quality graph or copy some unscaled diagram taken from the display on their graphics calculators. This still seemed to be the case and all such attempts lost some or all of the marks available. The axes should have been drawn with the aid of a ruler and pencil and, as a minimum, should have contained the appropriate scales; O(0, 0), (1, 0) and (2, 0) on the x-axis and (0, 0.5) and (0, 1) on the y-axis. The two concave arcs were often seen as being convex or as straight lines.

In part (b), there were many correct solutions with $\int_0^1 \frac{1}{2}(x^2+1)\mathrm{d}x = \frac{2}{3}$ often seen. Unfortunately, those candidates who drew straight lines in their sketches, usually ended up with P($X \le 1$) = $\frac{1}{2}(0.5+1) = \frac{3}{4}$; this gained no credit.

Part (c) was answered very well by many candidates with the standard of algebra and integration showing a marked improvement. However, it should be noted that, where an answer is given on the examination paper, sufficient method must be shown to ensure that full marks are awarded.

Part (d)(i) was answered very well by many candidates. However, some candidates, having found that Var(X) = $\frac{4}{5} - \left(\frac{19}{24}\right)^2 = \frac{499}{2880}$, then were unable to write down the value of k as 2880. Whilst most candidates obtained the correct answer in part (d)(ii), a minority thought incorrectly that E(X^2) = $\left(\frac{19}{24}\right)^2$ even though the correct value of $\frac{4}{5}$ was given in the question. Part(d)(iii) was quite well attempted although some candidates used Var(12X – 5) = 12 × Var(X) or even Var(12X – 5) = 12^2 × Var(X) – 5.

Q7.

Part (a) was answered very poorly by too many candidates. Candidates were asked to **state** values for the median and lower quartile of X. This should have indicated to candidates that the answers required little work, but this was often not picked up. Some candidates obtained the correct values but these were often invalidated by incorrect methods. Those candidates who employed calculus often had more success, as they realised that F(m) = 0.5 and F(q_1) = 0.25 were required.

Part (b) was often done well by those candidates who realised that F(x) = 1/54(x – 4)^3 + c, as they usually went on to use either F(1) = 1/2 or F(4) = 1 to show that c = 1. Some candidates, having been given the answer in the question, simply wrote this down at the end of what was usually a totally incorrect method.

Part (c) was usually done by correctly attempting to evaluate F(3) – F(2) but the arithmetic then defeated many.

Most candidates realised that F(q) = 3/4 was required and consequently went on to gain full marks for part (d)(i). The correct answer to part (d)(ii) was often seen, even by those candidates who could not do part (d)(i). However, some failed to give the answer to three decimal places and so lost the mark.

Q8.

On the whole, attempts at this question were rather disappointing. In part (a), when candidates did demonstrate that they knew that $F\left(\frac{3c}{4}\right) - F\left(-\frac{3c}{4}\right)$ was required, the algebraic manipulation then often defeated them. There were very few candidates who realised that the given F(x) was the cumulative distribution function for a *rectangular distribution*. Those who did recognise this fact were simply able to find the area of the rectangle of length $\frac{3c}{2}$ and height $\frac{1}{4c}$ to give the correct answer of $\frac{3}{8}$.

In part (b), although many candidates realised that they had to use the relationship f(x) = F′(x), they were then either unable to differentiate $\frac{x+c}{4c}$ with respect to x to give

$\frac{1}{4c}$ and/or neglected to explain what happens when they differentiate F(x) in the two regions $x < -c$ and $x > 3c$, simply stating the given answer did not suffice.

In part (c), although many candidates realised at this stage that they were dealing with a rectangular distribution and so used the appropriate formulae for E(X) and Var(X) from the table provided on page 11 of the booklet to obtain the required answers, many others wasted a lot of time on often fruitless attempts at integration. Also, candidates' inability to simplify E(X) = $\frac{1}{2}$ ($2c$) and Var(X) = $\frac{1}{12}(4c)^2$ resulted in a loss of marks.

Q9.

A number of candidates seemed to have the wrong perception that the request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram which they had managed to take from their graphics calculator display. All such attempts lost some or all of the marks available. The axes should ideally have been drawn with the aid of a ruler and pencil and should have contained the appropriate scales; O(0, 0), (2, 0) and (5, 0) on the x-axis and (0, 0.5) on the y-axis as a bare minimum. A concave curve from O(0, 0) to (2, 0.5) and a straight line from (2, 0.5) to (5, 0) should then have been drawn.

Most candidates realised that $\int_2^x \frac{1}{6}(5-x)\mathrm{d}x$ was required somewhere in part (b) even though some were unsure of the limits and what this integral gave them. Some thought incorrectly that F(x) = $\int \frac{1}{6}(5-x)\mathrm{d}x \Rightarrow \mathrm{F}(x) = -\frac{1}{12}(5-x)2^2 + c$, followed by F(5) = 1 $\Rightarrow c$ = 1, and hence that F(x) = 1 – $\frac{1}{12}$ $(5 - x)^2$ as required. This was very plausible but, unfortunately, the original assumption that F(x) = $\int \frac{1}{6}(5-x)\mathrm{d}x$ was false! However, that said, there were some excellent and fully correct solutions to part (b) where the correct expression F(x) = F(2) + $\int_2^x \frac{1}{6}(5-x)\mathrm{d}x$ or $\mathrm{F}(x) = \int_2^x \frac{1}{6}(5-x)\mathrm{d}x + k$ was used.

In part (c), the great majority of candidates were able to show that F(4) = $\frac{11}{12}$ and F(3) = $\frac{2}{3}$, and then went on to show that P(3 ≤ X ≤ 4) = $\frac{1}{4}$. However, some candidates treated this as if it were a discrete distribution, stating that P(X ≥ 3) = 1 – P(X ≤ 2). Unfortunately, all but the very best candidates assumed incorrectly that P(X ≥ 3 | X ≤ 4) = P(3 ≤ X ≤ 4) = $\frac{1}{4}$. Conditional probability was required using the fact that P(X ≥ 3 | X ≤ 4) = 1 –

$$\frac{\mathrm{F}(3)}{\mathrm{F}(4)} = 1 - \frac{8}{11} = \frac{3}{11} \text{ or } \mathrm{P}(X \geq 3 | X \leq 4) = \frac{\mathrm{F}(4) - \mathrm{F}(3)}{\mathrm{F}(4)} = \frac{1/4}{11/12} = \frac{3}{11}.$$

Q10.

In part(a), too many candidates seemed to have the wrong perception that the request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram which they had managed to take from the display on their graphics calculators. All such attempts lost some or all of the marks available. The axes and straight lines should have been drawn with the aid of a ruler and should have contained the appropriate scales: O or (0, 0), (1, 0) and (3, 0) on the x-axis and (0, 0.5) on the y-axis as a bare minimum. A straight line from (0, 0.5) to (1, 0.5) and a straight line from (1, 0.5) to (3, 0) should then have been drawn. It was often quite difficult to decide whether the sketch contained the required 'straight' lines or curves, so badly attempted were the sketches.

Similarly, part (b) was very poorly attempted with most candidates simply showing that f(1) = 0.5 or solving f(x) = 0.5 to give x = 1. Both of these were inappropriate and so gained no credit.

There were many good attempts at finding E(X) in part (c). However when, as here, the answer is given in the question, sufficient relevant working must be shown for full credit to be obtained.

Although there were many excellent solutions to part (d), there were only a few candidates who used the most efficient method of 1 – $\frac{1}{2} \times 0.75 \times \frac{3}{16}$; most chose an alternative method which required integration. There were a large number of usually less able candidates who wrote $\text{P}\left(X > 2\frac{1}{4}\right) = \text{f}\left(2\frac{1}{4}\right) = \frac{3}{16}$, thus gaining no marks.

Q11.

In general, the attempts at this question were very poor. It should be emphasised, yet again, that when an answer is given in a question, candidates must take extra care to explain what they are doing. This was often not seen in answers to parts (a)(ii), (a)(iii) and (b)(i), where many candidates showed insufficient working to achieve full credit.

Part (a)(i) asked candidates to sketch the graph of F. Many interpreted this as allowing the drawing of a rather scruffy freehand diagram without a scale. This was not what was intended. It was expected that scaled (approximately) and labelled axes be drawn (using a ruler and preferably a pencil) and care be taken in drawing the curved part of the sketch.

Although most candidates attempted the straight line from the origin to (1, ½) correctly, there was much less success with the curve from (1, ½) to (4, 1). The part of the graph defined by F(x) = 1 was often not drawn, and the part defined by F(x) = 0 was rarely seen at all.

Part (a)(ii) was not explained well by many candidates. Many thought incorrectly that dividing 4 (the length of the interval over which most of them drew their sketch) by 4 would give them the value of the lower quartile, whilst others assumed that (the value of the lower quartile) = $\frac{1}{2}$ × the value of the median) without any explanation as to why this was the case for this distribution.

It was, however, pleasing to see good candidates state correctly that $\text{F}(q_1) = \frac{1}{2}q_1 = 0.25$, leading to the given answer of $q_1 = \frac{1}{2}$. A variety of methods, mostly correct, were employed in part (a)(iii). The vast majority of candidates realised that the upper quartile was in the interval [1, 4] and used the fact that $F(q_3) = 0.75$ to good effect. Most used different rearrangements of the resultant cubic equation, whilst others used their calculators to show that $q_3 = 1.62$ (3 sf), to achieve the required given result.

The attempts at part (b)(i) were varied and mostly disappointing, with many candidates using the result $\alpha = \frac{1}{2}$ to show that $\beta = \frac{1}{18}$ without firstly explaining why they thought that $\alpha = \frac{1}{2}$. Some found only one relationship between α and β (usually either $\alpha = 9\beta$ or $\alpha + 9\beta = 1$) and then simply substituted the given values of α and β to show that these values did work. This was not felt to be sufficient to gain full credit since each of these equations, taken in isolation, had infinitely many solutions, of which the given solution was just one such pair of values. However, there were a few excellent solutions, the most successful of these first showing that on the interval [0, 1], $f(x) = \alpha = F'(x) = \frac{1}{2}$. This was then followed by use of the property that f was continuous at $x = 1$ to give $f(1) = \alpha = 9\beta$, thus giving $\beta = \frac{1}{18}$ as required.

Part (b)(ii) was a good source of marks, even for those candidates who gained virtually no credit on earlier parts of the question. These candidates showed good algebraic and integration skills in finding correctly that $E(X) = 1.125$. A small minority of candidates, instead of using any integration methods, used their calculators to find the correct numerical answer, usually after stating correctly that $E(X) = \int_0^1 \frac{1}{2}x\,dx + \int_1^4 \frac{1}{18}x(x-4)^2\,dx$. This method gained the full 5 marks for those candidates who stated the correct numerical value but only 1 mark for those who did not. Whilst the use of calculators is encouraged in data analysis and in the evaluation of numerical expressions, candidates run the risk of losing marks if they acquire an incorrect numerical answer without any intermediate working.

Q12.

In part (a), many candidates were able to sketch the graph of f correctly. However, 'sketch the graph' does not give candidates licence to produce rough, freehand diagrams devoid of scales. In many cases, it was very difficult to give candidates credit since it was impossible to infer whether they intended to draw straight lines or curves. Consequently, some candidates lost some or all the marks for this part of the question. In part (b)(i), many candidates used $\int_0^2 \frac{2}{15}t\,dt$, usually gaining the correct answer of $\frac{4}{15}$. However, as the graph contained straight lines, integration, although correct, was not the most efficient method. Although there were many correct solutions seen to part (b)(ii) by a variety of correct methods, there were still some candidates who treated $f(t)$ as a discrete distribution and consequently thought incorrectly that $P(2 < T < 4) = P(T \le 3) - P(T \le 2)$. This method gained no credit. Some candidates, presumably from experience on past papers, first derived the cumulative distribution function, $F(t)$. Although this method gained the better candidates full marks, it was not the most efficient way of doing this part of the question. In part (c), the great majority of candidates considered the correct integrals and performed correct integration and were then able to evaluate these using correct limits to gain the required answer of $2\frac{2}{3}$. However, some candidates were unable to substitute the limits correctly in order to obtain this correct answer. Unfortunately, those candidates who thought f (t) to be discrete earlier in the question continued with this incorrect premise here.

Q13.

In part (a), F(0) = $\frac{1}{k+1}$ was all that was required, but the use of F(0) – F(–1) = $\frac{1}{k+1}$ was not penalised. In part (b), most candidates realised that they needed to use the fact that F(q_1) = 0.25 and so went on to determine the correct expression for q_1. Unfortunately many candidates must have thought that k was the lower quartile since they found an expression for k in terms of x. In part (c), candidates were expected to use the relationship f(x) = $\frac{d}{dx}(F(x))$ in order to gain the required given answer. Unfortunately, attempts at differentiating F(x) = $\frac{x+1}{k+1}$ were, in the main, unconvincing with little or no attention paid to F(x) = 0 and F(x) = 1. In part (d)(i), most candidates gained full marks. Many candidates used integration methods in part (d)(ii) which, although correctly done, did waste a lot of time, especially as the answer for E(X) = 5 could have been written down by simply considering the sketch drawn in part (d)(i). Similarly, Var(X) = $\frac{1}{12}$ (11 – –1)^2 was a more efficient method of finding the variance. There were many well-explained solutions to part (d)(iii) with most candidates realising that P(2 < X < 5) was required. Many candidates correctly stated that P(X < q_1) = 0.25 but then assumed that P(X < E(X)) = 0.5 without any explanation. This statement was true here since, for a rectangular distribution, the median and the mean take the same value.

Q14.

Most candidates, in part (a), made a good attempt at sketching the graph of the probability function, although there were the usual weak attempts where candidates did not recognise that f(t) = $\frac{3}{8}t^2$ is a quadratic curve whilst f(t) = $\frac{1}{16}(t+5)$ is a straight line.

Part (b) was rarely attempted using the most efficient method of finding the area of a trapezium, but those candidates who did usually did so correctly. The vast majority attempted to use integration when answering this part of the question, with either $\int_1^3 \frac{1}{16}(t+5)\,dt$ or $1 - \int_0^1 \frac{3}{8}t^2\,dt$ often being used correctly. However, integration was not necessary and its use here again used up valuable time which could have been helpful to some candidates elsewhere. It was disappointing to see that only the most able students coped well with part (c)(i). Very few realised that F(t) = F(1) + $\int_1^t f(t)\,dt$ was required here, with the vast majority simply evaluating $\int_1^t \frac{1}{16}(t+5)\,dt$ and then writing down the answer that was given in the question. Part (c)(ii) was tackled much better, with many candidates obtaining the correct answer. The negative root to the obtained quadratic equation was correctly rejected but often without any justification being given.

Q15.

There were very many excellent solutions to this question with full marks seen on a regular basis; however some candidates seem to struggle with questions where the cumulative distribution function is given, rather than the probability density function. In part (a), some candidates were unable to differentiate the function and some treated the given function as the probability density function and attempted, usually with very little success, some sort of integration. Of the very many candidates who did manage to differentiate F(x) correctly, a few failed to quote the full answer for f(x), simply giving the answer as $\frac{1}{9}$.

In part (b), those candidates who had obtained the correct answer to part (a) usually drew the correct graph. Unfortunately, some candidates showed their lack of understanding of what was given in the question by simply attempting to draw the graph of F(x). Part (c) was done well by those candidates who understood the question, many using the graph that they had drawn in part (b) to help them to obtain the correct answer. For those who realised that this was a continuous rectangular distribution, part (d) was done very quickly by using the correct formulae from page 11 of the supplied booklet of formulae and statistical tables. However, some candidates attempted, usually with little success, to use integration methods in order to find the mean and variance of the given distribution.

Q16.

Part (a) caused the most problems in this question. Although many candidates were able to demonstrate why f(x) =10 for $-0.5 \le x \le 0.5$, they were unable to indicate why the values of X had to lie in the given range. Part (b) was often done correctly, if not always by the most efficient method. For this rectangular distribution, there were far too many candidates still using integration, rather than a sketch diagram and simple geometry, to reach the required result. It was expected that candidates would use the formulae from page 11 of the booklet provided to answer part (c). Although many did, there were those who attempted integration methods with little success. Some candidates who managed to find the variance then did not go on to find the standard deviation.

Q17.

Part (a)(i) was usually done well, with most making sufficient reference to the area of a rectangle to gain the mark. Some candidates insisted on using calculus in this topic and, although they usually went on to show the given result, it did mean that more time was spent on this part of the question than was anticipated. In part (a)(ii), most candidates were able to show that E (X) = $\int_a^b kx\,dx = \left[\frac{kx^2}{2}\right]_a^b = \frac{1}{2}k\left(b^2 - a^2\right)$. Unfortunately, many then simply wrote down the given answer without any further evidence of method. Others often seemed unfamiliar with the method of 'the difference of two squares' in the factorisation of $b^2 - a^2$ and so simply tried to fudge the given answer. In part (b)(i), the answer $\mu = 1$ was usually stated correctly. In part (b)(ii), although the majority of candidates used the formula $\sigma^2 = \frac{1}{12}(b-a)^2$, quoted from the formula booklet provided, to show that $\sigma = \sqrt{3}$, there were several who used integration to find E(X^2) before using Var (X) = E(X^2) – μ^2, usually correctly. Unfortunately, some of the mainly weaker candidates thought that $\sigma = \frac{1}{12}(b-a) = 3$. With follow–through marks being available in part (b)(iii), many candidates were able to show a good understanding of the rectangular distribution. However, some candidates again used calculus unnecessarily to work out the given probability, with $\int_{-2}^{1/\sqrt{3}} \frac{1}{6}\,dx$ often seen and correctly evaluated.

Q18.

The use of calculus in questions of this type still seemed to be a popular, yet unnecessary, method for a small minority of candidates. Most were able to find the correct value of a in part (a), although it did defeat more candidates than expected. Part (b) was usually done well by those who used the correct formulae from the booklet. In part (c), many candidates failed to understand the concept of the magnitude of the error.

Q19.

In part (a), candidates showed poor ability in sketching graphs. Common faults were the omission of essential values on both axes, lines drawn as incorrectly passing through the origin on the interval [0 , 1] and the quadratic curve incorrectly drawn as convex on the interval [1 , 4]. In part (b)(i), the vast majority of candidates either ignored the required limits of 0 and x altogether, or used limits of 0 and 1. Consequently, although most stated the required answer, many lost a mark. Part (b)(ii) was usually completed correctly with the correct answer of 0.85 often seen. Part (b)(iii) was not done well. Very few candidates tried to 'verify' the given result. Instead, most attempted to calculate q_x so that they could then show that its value fell within the given interval.

Q20.

Part (a)(i) was answered well by all but the very weakest candidates. However, there were still some candidates who used integration to achieve the required result instead of using $E(X) = \frac{1}{2}(a + b)$ from the formulae booklet. There were many very pleasing solutions seen to part (a)(ii). Part (b) was usually done best by those candidates who used a diagram to demonstrate their understanding of what was required.

Q21.

In part (a), candidates showed poor ability in sketching graphs. Common faults were the omission of essential values on both axes, lines drawn as incorrectly passing through the origin on the interval [0 , 1] and the quadratic curve incorrectly drawn as convex on the interval [1 , 4]. In part (b)(i), the vast majority of candidates either ignored the required limits of 0 and x altogether, or used limits of 0 and 1.

Consequently, although most stated the required answer, many lost a mark. Part (b)(ii) was usually completed correctly. Several candidates wasted time in part (b)(iii) by considering the probability density function for $x > 1$ with the inevitable loss of most, if not all, of the available marks. Those candidates who did obtain a quadratic equation usually managed to solve it correctly and nearly always realised that one of their solutions was inadmissible. Part (b)(iv) was often done well, even by those candidates who failed to appreciate what was required in part (b)(iii).

Q22.

Statistics 2A

In general, candidates seemed to have less success with this question. In part (a), the correct graph was only seen very infrequently, with the majority of candidates drawing straight lines to represent both parts of the graph. Where a curve was drawn from $t = 3$ to $t = 6$, it was often concave rather than convex. The vast majority of candidates correctly gave the answer of 'zero' to part (b). However, a minority treated this as a discrete distribution and used P ($X = 3$) = P ($X \leq 3$) – P ($X \leq 2$) = $\frac{1}{5}$. Part (c) was usually done well. However, many candidates used the more difficult integration route and, the correct answer of 0.4 was often seen. Part (d) was usually attempted quite well with most candidates aware that the median value, m, had to satisfy $\int_0^m \frac{1}{5}\,dt$ = 0.5. Part (e) was found difficult by all but the more able candidates. Calculating the mean to be 2.55 using the integration of two functions for the two different ranges was beyond most candidates, with several attempting to integrate one or other of the functions between 0 and 6. The final part, calculating P(median < T < mean) was beyond almost all but the best candidates.

Statistics 2B

In general, candidates seemed to have less success with this question than others on the paper. In part (a), the correct graph was seen very infrequently, with the majority of candidates drawing straight lines to represent both parts of the graph. Where a curve was drawn from $t = 3$ to $t = 6$, it was often concave rather than convex in nature. The great majority of candidates correctly gave the answer of 'zero' to part (b). However, a minority treated this as a discrete distribution and used P ($X = 3$) = P ($X \leq 3$)– P ($X \leq 2$) = $\frac{1}{5}$. Part (c) was usually done well. However, many candidates used the more difficult integration route and, the correct answer of 0.4 was often seen. Part (d) was usually attempted quite well with most candidates aware that the median value, m, had to satisfy $\int_0^m \frac{1}{5}\,dt$ = 0.5. Part (e) was found difficult by all but the more able candidates. Calculating the mean to be 2.55 using the integration of two functions for the two different ranges was beyond most candidates, with several attempting to integrate one or other of the functions between 0 and 6. The final part, calculating P(median < T < mean) was beyond almost all but the best candidates.

Q23.

Candidates should be given more information regarding the terminology that is used in questions. Where **'state'** or **'write down'** are used it is to indicate to candidates that the answer can reasonably be simply written down and that working does not need to be shown. There were a great number of scripts seen where integration was used in order to find the answers to parts (a) and (b). Although the correct answer to each of these parts was eventually arrived at, it was certainly not the most efficient way of doing so. The value of P (X >0) was usually found correctly to be 0.6 but again, far too many candidates wasted time by using integration. Part (d) was only usually done correctly by the most able candidates. The vast majority of candidates usually perceived incorrectly that P (X >3.5) was required and so did not attempt to find P (X < – 3.5) + P (X > 3.5).

Q24.

Part (a) was generally done very well with many candidates producing a reasonably correct sketch even if by no means to scale!. Whilst there were occasional omissions, some candidates found trouble sketching the graph of f(x) = c, let alone the second part of the graph. Again, part (b) was done well by most candidates. Even those candidates with a deficient graph made some use of the fact that the total area was 1 unit. Sadly, in part (c), the vast majority of candidates did not convert 0.01 metres to 10 mm; were they careless or confounded? Consequently their area was totally in the rectangular section of the graph. Those candidates who considered $P(X < 10)$ invariably were successful.

Q25.

The responses to this question were very disappointing, with the great majority of candidates unable to distinguish between the distribution function, $F(x)$ as given, and the probability density function $f(x)$. Part (a) was usually only answered correctly by the most able candidates. Instead of simply evaluating $F(3) - F(2)$, the great majority integrated $\frac{x^3}{64}$ before substituting values of $x = 3$ and $x = 2$. Part (b) was rarely attempted correctly with differentiation of $F(x)$ to find the required form for $f(x)$ $= \frac{3x^2}{64}$ infrequently seen. In part (c)(i), most candidates assumed that $f(x) = \frac{x^3}{64}$, again revealing an inability to distinguish between the probability density function and the distribution function. In part (c)(ii), too many candidates only attempted to work out $\int x^2 f(x)\,dx$, and did not then subtract $[E(X)]^2$ in order to find Var(X).

Q26.

This proved a difficult question on which to score full marks. Many candidates scored full marks in part (a) though a small proportion lost marks through simple arithmetic errors. However, full marks in part (b) were rare indeed. Most of the many unsuccessful attempts centred on the incorrect use of formulae from the supplied booklet. Correct attempts usually first involved finding the value of a scaling factor so that the conditional area above 5 minutes was equal to unity.

Q27.

Here again a wide range of marks was seen. All but the weakest candidates found part (a) accessible and, by substituting a and $a + k$ into formulae from the booklet, were able to score four marks. Whilst many candidates scored the two marks in part (b), a significant proportion carelessly equated [(–6 + 24) – (–6)] to 30.

Q28.

Most candidates knew how to use the distribution function to calculate probabilities and so parts (a) and (b)(i) caused few problems. However, there was some confusion in answering part (b)(ii). Some candidates were unsure which probability they were required to find whilst a few attempted to multiply probabilities for events that were not independent. It was pleasing to note that part (c) was generally well answered.

Q29.

Many candidates scored full marks on this first question. Part (a) proved to be accessible to all candidates. In part (b) a majority of candidates assumed $s = 10$, then found that the total area was 0.3 + 0.2 + 0.5 = 1. However, about 50% of these candidates lost one mark for not stating clearly that an area of 1 is a requirement for a probability density function.

Those candidates who equated an area in terms of s to 1 found the solution more difficult. However, many candidates eventually showed that, as a result, 0.05s must equal 0.5, or an equivalent equality.

Q30.

More able candidates scored 10 marks on this question but others only scored about half this number.

In part (a) knowledge of the formulae for mean and variance was good. However about half of the candidates then showed that a = 12 and b = 30 satisfied the requirements; this was not considered a proof. More able candidates set up and then solved two simple simultaneous equations, but almost all failed to realise that $\sqrt{324}$ = ± 18 and lost one of the five marks available.

Parts (b) and (c) proved to be very accessible to many candidates. However, some failed to recognise, in part (b), that the lower limit was 12 so that the height was $\frac{1}{18}$, not $\frac{1}{30}$. This error was carried forward into part (c), although P(X < 16.5) was often deduced from substitution in the given algebraic expression. A minority of candidates, particularly in part (b), attempted to use a normal distribution, for which no marks were awarded.

Q31.

This question proved to be a good source of marks for many candidates. In part (a)(i), most candidates used integration, with both $\int_6^{7.5}\left(2-\frac{4}{15}t\right)$ dt and 1 – $\int_3^6 \frac{1}{90}t^2$ dt used and correctly evaluated. Only a few candidates, who had initially sketched the probability density function, used the area of a triangle in order to gain the given answer of 0.3.

There were many excellent attempts at part (a)(ii) with most candidates, realising that the median value had to lie between 3 and 6, correctly used $\int_3^m \frac{1}{90}t^2$ dt =0.5. Unfortunately, some candidates used zero as the lower limit and consequently gained a value of m = 6.46, this being outside the range of values of t for which the function f (t) = $\frac{1}{90}t^2$ is defined.

Part (b) was usually very well done with the vast majority of candidates realising that $\int t\,\mathrm{f}(t)\,\mathrm{d}t$ had to be considered. Most proceeded to attempt to evaluate $\int_3^6 \frac{1}{90}t^3\,\mathrm{d}t$ + $\int_6^{7.5}\left(2t-\frac{4}{15}t^2\right)$ dt. Some candidates, unfortunately, tried to evaluate one or both of these integrals using the limits 3 and 7.5.

Q32.

In part (a) the large majority of sketches scored full marks. Where marks were lost it was usually for an incorrect labelling of the vertical axis or ignoring one part of f(x). The latter often resulted in a triangle in the interval 0 to 60. Correct graphs were usually followed by correct answers to parts (b) and (c). Candidates with incorrect graphs made little headway in part (b) but sometimes recovered in part (c) by calculating 1 – (area of triangle from 0 to 30).

Q33.

In part (a) correct diagrams having axes marked with the pertinent values were the norm.

In part (b) $\frac{1}{54}t^3$ was frequently seen but only the most able candidates presented fully correct solutions. The vast majority of candidates simply did not know how to cope with finding F(t) when f(t) was defined in two intervals.

Part (c) was answered very well and by a variety of methods with the majority of candidates gaining full marks.

Q34.

Almost without exception, candidates scored the first eight marks. In part (b)(iii), the conditional probability caused many candidates problems. They simply evaluated P(X < 12)or quoted 0.80 × 0.65. However a significant proportion of candidates, from across all grades, scored full marks by either using $\frac{0.65}{0.80}$ directly, or by finding that, for the (conditional) distribution between 4 and 18, $c = \frac{1}{16}$ and thus P(x < 12)= $\frac{13}{16}$. Many candidates scored the mark in part (c) usually by indicating that delays may exceed 18 minutes.

Q35.

It was expected that most candidates would find part (a)(i) very accessible. As stated earlier in the report, this was not the case due to poor algebraic skills. As a result, many candidates scored at most one of the two marks in part (a)(ii) for a correct method, not for a fiddled answer. Again, in part (b), many candidates lost the mark because they were unable to solve $19c^2 = 171$ correctly. In part (d)(i), even with follow-through marks, candidates simply could not find the 'correct' probability; some even tried using a normal distribution. In part (d)(ii), many candidates simply evaluated (7c – c) × 0.25, apparently forgetting that the distribution started at c. Having said all this, some candidates scored all eleven marks through a good knowledge of the relevant statistics and the ability to handle relatively simple algebra.

Q36.

This question was well answered with most candidates scoring most of the marks available.

In part (a), sketches were usually adequate even if a little rough at times. Some candidates lost a mark for labelling the point 2.5k on the vertical axis as 2.5. In part (b), correct geometrical reasoning (common) or integration (uncommon) scored full marks. Similarly, in part (c), the chosen method was invariably fully correct.

Q37.

It was disappointing to see the large number of incorrect answers to part (a); this despite the relevant formulae being listed on the supplied booklet. Additionally, candidates often simplified $\frac{(4c)^2}{12}$ to $\frac{4c^2}{12}$ or $\frac{16c}{12}$.

In general, only the best candidates made any headway in part (b), but even some of these were hampered in using Var (X) = E(X^2) – (E(X))^2 by their algebraic errors in part (a). A small number of candidates equated $\frac{28}{3}$ to $\int_{-c}^{3c} \frac{x^2}{4c}\,dx$,often with complete success.

Answers to part (c)(i) were almost always correct and it was pleasing to see good attempts at calculating the probability in part (c)(ii). However, although most associated diagrams showed a lower limit of –2, many candidates apparently ignored the modulus signs and calculated P(0 < X < 2.5).

Q38.

In part (a) it was very disappointing to see P(T = 4) = 0.25 in a large number of scripts. The correct answer of 'zero' was seen very infrequently.

In part (b) there were many candidates who correctly considered $\int_0^m \frac{t^2}{18}\,dt$ = 0.5 and usually went on to gain full marks. However there were many other unsuccessful attempts to show that the median was 3. These mainly consisted of the substitution of t = 3 into one of the expressions $\frac{t^2}{18}$ or $\frac{1}{4}(5-t)$, giving an answer of 0.5. This then led to a false deduction.

Many very good solutions were seen to part (c) despite the indifferent starts seen in parts (a) and (b). The most common approach was by integration methods with very few candidates using the geometry of the distribution to achieve the required result. Unfortunately, a few candidates thought that <3.5 meant that the limit to use was 3.45 or 3.49.

Many excellent attempts at the correct method to find the mean were seen in part (d). Unfortunately, some candidates could not cope with the fairly simple and straightforward integration.

Q39.

This proved to be a good starter question for most candidates. In part (a), all candidates were aware that the total area had to be unity and finding this area, in terms of c, was undertaken in a variety of ways. Whilst most attempts were correct, some were confused or unclear and the solutions would have benefited from a sketch illustrating the method. Many candidates used symmetry to find the total area but few candidates equated half the area to 0.5. Similarly, very few candidates assumed c = 0.01 and then showed that the total area was unity. The few candidates who simply wrote 100c = 1 so c = 0.01 scored at most one mark. Most candidates scored full marks in part (b). Of those who did not, most candidates used incorrect heights, particularly at X = 20, or found the wrong area, for example P(X > 10).

Q40.

The majority of candidates showed a good understanding of this topic and so were able to score a good proportion of the marks available.

There were many very good sketches seen in part (a), although some candidates thought that $\frac{1}{54}t^2$ was the equation of a straight line. Parts (b)(i) and (ii) were well done by those candidates who used the areas of a rectangle and triangle to obtain their answers. Despite candidates being instructed to use their graphs in this part of the question, some candidates still used integration methods and so almost certainly wasted valuable time.

Part (b)(iii) was often attempted by use of $\int_0^m f(t)\,dt = 0.25$. Unfortunately, most candidates wrongly thought that this equated to the use of $\int_0^m \left(\frac{t^2}{54}\right) dt = 0.25$, and consequently gained no credit. The correct answer of m = 3.5 was rarely seen. There were many excellent solutions to part (c) with the correct answer of $\frac{4}{81}$ being seen on the majority of scripts.

Q41.

Most candidates made a good start with this question and the results for the mean and standard deviation of the exponential distribution were well known. The majority of candidates used the given formula to calculate the probability in part (b) and to find the median in part (c). A few attempted both these parts by integrating the function they had written down in part (a)(i) and this wasted time.

Q42.

The many candidates who scored full marks for their sketches in part (a) then usually went on to score all the marks for their solutions to parts (b) and (c). Candidates whose sketches were incorrect, often through ignoring the range $0 \le x \le 1$, fared less well. In part (b) they equated a triangular area of $\left(\frac{1}{2} \times 2 \times c\right)$ to 1, giving c = 1. In part (c), such candidates then calculated a trapezium area of $\left(\frac{1}{2} \times \left(c + \frac{c}{2}\right) \times 1\right) = \frac{3}{4}$. These incorrect simplified solutions generally scored two out of the six marks available.

Q43.

In part (a), a significant number of candidates used a and k rather than a and $a + k$, or did not simplify their correct expressions. Those candidates who obtained correct simplified expressions in part (a) generally scored the three marks in part (b). Here about half the solutions involved solving 2a + k = 42 and k^2= 36 for k and a, while the others showed that a = 18 and k = 6 satisfied E(X) = 21 and Var(X) = 3. Correct answers to part (c) were the norm, with most candidates recognising that the answer equated to the rectangular area of (24 – 20) x $\frac{1}{6}$.

Q44.

A majority of candidates scored either 7 or 8 marks on this question. In answering part (a), most candidates showed sufficient working to show they had obtained 23.5c (areas of rectangles, triangles, trapezia or simple counts) and equated it to unity. A proportion of candidates stated correctly the value of zero for part (b) but many stated the value of f(250). Most candidates recognised the area that was required in part (c) and many evaluated it correctly, though some tried to find it as a single trapezium. A smaller number of candidates used f(249) + ... + f(252).

Q45.

Most candidates scored the 2 marks in part (a), though a few chose to divide by 2 rather than 12 when finding the variance. It was pleasing to see the large number of correct answers to parts (b) and (c)(i); the usual incorrect values were $\frac{1}{6}$ or $\frac{1}{7}$. In answering part (c)(ii), candidates tended to use their value of σ^2, rather than σ and/or failed to find an actual area by not using their answer to part (b). Those candidates who sketched a diagram were often the more successful.

Q46.

Many candidates scored full marks on this question. In part (a), there were many correct sketches though some candidates started their vertical axis at $5k$, rather than at 0: this caused them difficulties throughout the question. In answering parts (b) and (c), most candidates either used geometry or integration, though a few used a mixture. Whilst errors were rare, those candidates who used rectangles and triangles were generally less successful than those who used trapeziums and, in integration, incorrect limits were sometimes in evidence. There were few fudged answers to part (b).

Q47.

In part (a) there were many excellent solutions, with the vast majority of candidates able to manipulate fractions to obtain the correct value of k . Unfortunately some candidates used inexact decimal calculations, which was unacceptable at this level when an exact value was required. Since the answer was given in this question a full and detailed method was essential for full marks to be awarded. Although there were many excellent solutions to part (b), several candidates wrongly thought that P(S < 3) = $\int_0^3 f(s)ds$ where $\int_1^3 f(s)ds$ was required. Those candidates who attempted to find the distribution function invariably omitted the constant of integration. The correct form of the distribution function was:

$$F(s) = \begin{cases} 0 & s < 1 \\ \frac{1}{124}(s^3 - 1) & 1 \le s \le 5 \\ 1 & s > 5 \end{cases}$$

This then gave F(3) = $\frac{26}{124} = \frac{13}{62} = 0.210$.

Part (c) was not well attempted, with many candidates confusing the median with the mean and consequently evaluating $\int_1^5 sf(s)ds$ instead of using $\int_1^m f(s)ds$ = 0.5, giving m^3 = 63 with a final answer of 4 minutes. However, some of those following the correct method lost an accuracy mark by not giving the answer correct to the nearest minute. As with part (b), some candidates thought that the lower limit was zero and consequently lost marks.

Q48.

Attempts at this question on a rectangular distribution were much improved over answers on previous papers. In part (a), almost all candidates reasoned correctly as to why $k = 12$ and then scored the 3 marks in part (b), though a minority used $\sigma^2 = \frac{1}{12}(b - a) = 1$ or $\sigma = 12$.

Answers to part (c) (i) were sound and, in the main, scored full marks. However, in part (c) (ii), many candidates used $\frac{(\mu - \sigma) - 500}{12}$, often obtaining the correct value of 0.211, but then did not find the required answer of 1 – 0.211 = 0.789.

Q49.

In the main, sketches were adequate, though some candidates almost replicated the sketch given in Question 5. Adequate sketches resulted in many candidates providing a satisfactory proof in part (b) and a subsequent correct answer to part (c).

Very few candidates scored the 5 marks available in part (d). Whilst many candidates realised that P(2 < H <2½) was required and then obtained the correct value of $\frac{5}{24}$ very few then moved to the necessary conditional probability. A-small number of candidates found a new value of c, namely $\frac{1}{2}$, so that $P(2 < H < 3) = 1$ and then evaluated $P(2 < H < 2\frac{1}{2}) = \frac{5c}{4} = \frac{5}{8}$ for full marks.

Q50.

Although this was not a particularly high scoring question, candidates as a body seemed to prefer it to questions set on previous papers which have required integration. A few candidates were unaware that $\int_1^2 (x - 1.54)^2 \, f(x)\,dx$ is the variance. Instead they expanded this integral and hence evaluated $E(X^2)$ for part (b)(iii). They then used this to evaluate the standard deviation in part (b)(ii).

Q51.

Some candidates were thrown by this question which was in an unfamiliar form. Explaining why 0.75 was the mode caused particular problems. However, the stronger candidates scored well.

Q52.

Most candidates used integration throughout this question. Apart from part (c)(i), methods based on diagrams or quoting formulae were equally acceptable. Part (d) caused the most problems, with some apparently unsure about the meaning of 'magnitude'.

Q53.

Some found this question easy but many were put off by the integration. Integration was only needed for part (a)(ii). Part (a)(i) required no proof; part (b) could be answered using a diagram and part (c) by using the earlier results. However, those who were unable to deal with the integration were generally put off from attempting the rest of the question.